



MÁSTER EN FINANZAS CUANTITATIVAS



GUÍA DIDÁCTICA Módulo 9

CURSO 2025/2026

Módulo 9: Gestión de Riesgos Financieros

2025/2026

Coordinador:

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UNED. Fac. de Ciencias Económicas y Empresariales.



Curso 2025/2026



MÁSTER EN FINANZAS CUANTITATIVAS **CURSO 2025/26**

PRESENTACION

El Máster en Finanzas Cuantitativa se integra por 10 módulos que se agrupan en los siguientes tres cursos de experto.

Curso de Experto en Métodos Computacionales en Finanzas

Módulo 1: Programación en Python y en R

Módulo 2: Modelos de Simulación

Módulo 3: Datos Financieros y Aprendizaje Automático

Curso de Experto en Métodos Econométricos Aplicados en Finanzas

Módulo 4: Modelos de Series Temporales

Módulo 5: Modelos de Volatilidad y Correlación

Módulo 6: Modelos de No linealidad

Curso de Experto en Análisis Financiero y Gestión de Riesgos

Módulo 7: Valoración de Instrumentos Financieros

Módulo 8: Gestión de Carteras de Inversión

Módulo 9: Gestión de Riesgos Financieros

Módulo 10: Teoría del Valor Extremo

Atención directa al alumno:

Prof. Manuel Sánchez Sánchez

UNED

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Guía de conceptos matemáticos y estadísticos para la Gestión del Riesgo Financiero

Topics

- Course Overview
- Risk Management Overview
- Some Basic Math
 - ▣ Returns
 - ▣ Discount factors
 - ▣ Geometric Series
 - ▣ Combinatorics

Course Overview

1. Some Basic Math
2. Probabilities
3. Basic Statistics
4. Distributions
5. Multivariate Distributions & Copulas
6. Bayesian Analysis
7. Hypothesis Testing & Confidence Intervals
8. Matrix Algebra
9. Vector Spaces
10. Linear Regression Analysis
11. Time Series Models
12. Decay Factors

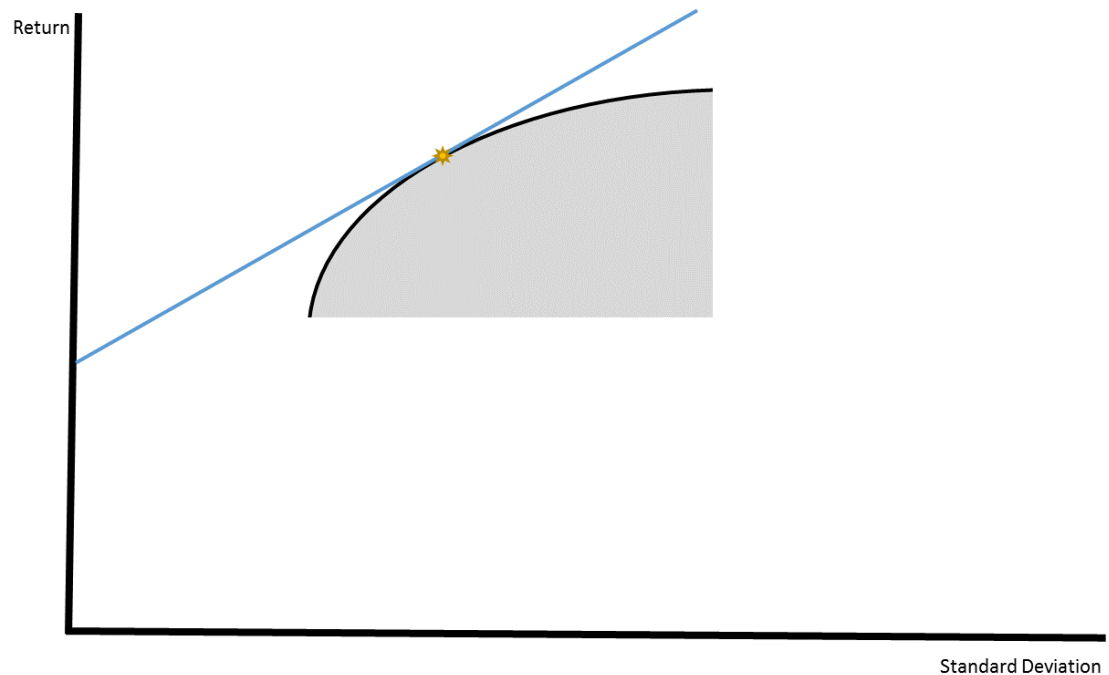
Risk Management Overview

A Brief History of Quantitative Financial Risk Management: Modern Portfolio Theory (MPT)



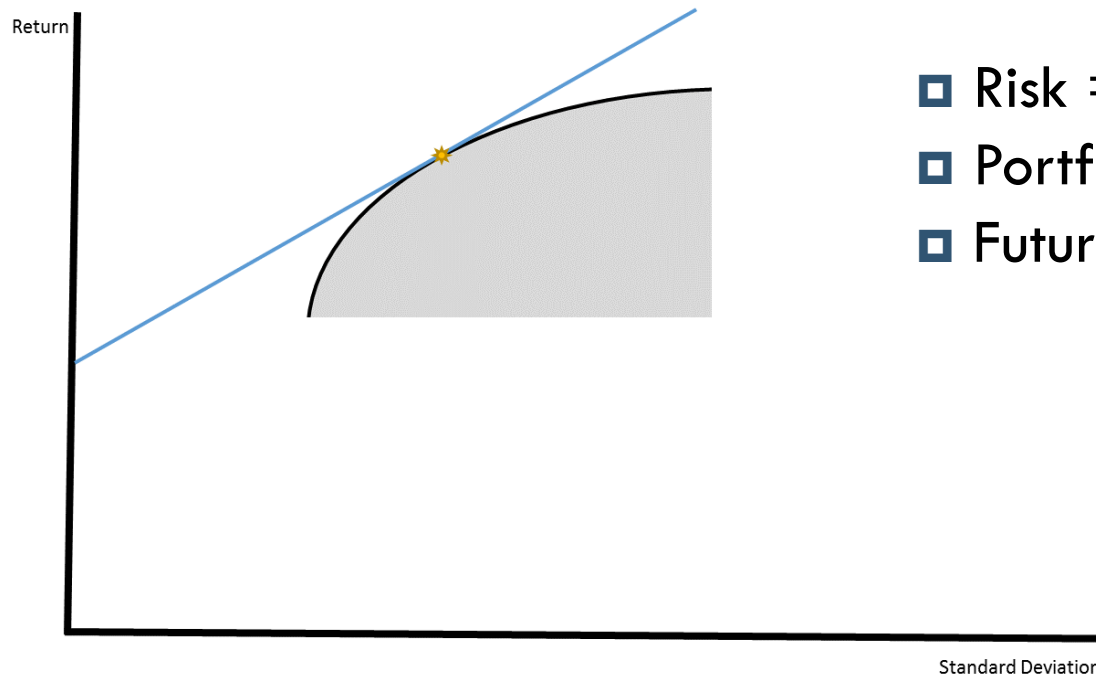
Harry Markowitz

Source: inversionario.com



Risk Management Overview

A Brief History of Quantitative Financial Risk Management: Modern Portfolio Theory (MPT)



- Risk = Standard Deviation
- Portfolio Risk = Correlation
- Future = Past

Returns

Log Returns

- $r_t = \ln(1 + R_t)$

- $r_t \neq \ln(R_t)$

- $r_t = \ln\left(\frac{P_t}{P_{t-1}}\right) = \ln(P_t) - \ln(P_{t-1}) = p_t - p_{t-1}$

- $R_{n,t} = (1 + R_{1,t})(1 + R_{1,t-1}) \cdots (1 + R_{1,t-n+1}) - 1$
 $r_{n,t} = r_{1,t} + r_{1,t-1} + \cdots r_{1,t-n+1}$

Returns

Continuously Compounded Returns

$$\square R_{\text{Annual}} = \left(1 + \frac{R_{\text{Nominal}}}{n}\right)^n - 1$$

$$\square \lim_{n \rightarrow \infty} R_{\text{Nominal}} = r$$

Discount Factors

- Present value of V_{t+n}

$$V_t = \frac{V_{t+n}}{(1+R)^n}$$

$$V_t = \left(\frac{1}{1+R} \right)^n V_{t+n} = \delta^n V_{t+n}$$

Geometric Series

□ Infinite Series

$$S = \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \dots$$

$$S = \sum_{i=1}^{\infty} \delta^i$$

$$S = \sum_{i=0}^{\infty} \delta^i = \frac{1}{1-\delta}$$

Geometric Series

□ Finite Series

$$S = \sum_{i=0}^{n-1} \delta^i$$

$$\delta S = \sum_{i=0}^{n-1} \delta^{i+1} = S - \delta^0 + \delta^n$$

$$S = \frac{1 - \delta^n}{1 - \delta}$$

Combinatorics

□ n objects, choose k :

$$C(n, k) = \binom{n}{k} = \frac{n!}{k!(n-k)!}$$

Mathematics & Statistics for Financial Risk Management

2: Probabilities

Topics



- Discrete Random Variables
- Continuous Random Variables
- Mutually Exclusive Events
- Independent Events
- Probability Matrices
- Conditional Probability

Discrete Random Variables

Variables can take on only a countable number of values.

$$P[X = x_i] = p_i \text{ s.t. } x_i \in \{x_1, x_2, \dots, x_n\}$$

$$\sum_{i=1}^n p_i = 1$$

Continuous Random Variables

- Variable can take on any value within a given range.

$$P[r_1 < X < r_2] = p$$

- PDF:

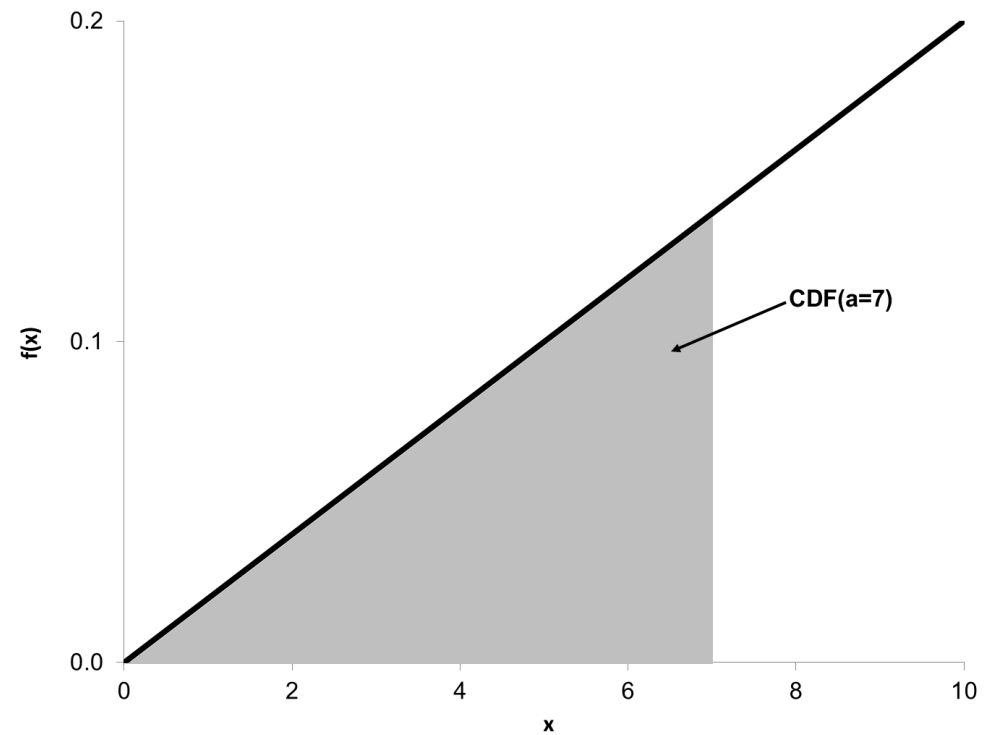
$$\int_{r_1}^{r_2} f(x) dx = p$$

$$\int_{r_{min}}^{r_{max}} f(x) dx = 1$$

Continuous Random Variable

□ CDF

$$F(a) = \int_{-\infty}^a f(x) dx = P[X \leq a]$$



Mutually Exclusive Events



$$P[A \cup B] = P[A] + P[B]$$

Independent Events



$$P[\text{rain and market up}] = P[\text{rain} \cap \text{market up}] = P[\text{rain}] \cdot P[\text{market up}]$$

Probability Matrices

		Stock		
		Outperform	Underperform	
Bonds	Upgrade	15%	5%	20%
	No Change	30%	25%	55%
	Downgrade	5%	20%	25%
		50%	50%	100%

Conditional Probability

$$P[\text{market up} \mid \text{rain}] = p$$

- Relationship to unconditional probability:

$$P[\text{market up}] = P[\text{market up} \mid \text{rain}] \cdot P[\text{rain}] + P[\text{market up} \mid \overline{\text{rain}}] \cdot P[\overline{\text{rain}}]$$

Mathematics & Statistics for Financial Risk Management

Topics

- Averages
- Expectations
- Variance & Standard Deviation
- Standardized Variables
- Covariance & Correlation
- Hedging
- Moments
- Skewness & Kurtosis
- Coskewness & Cokurtosis
- BLUE

Averages

- Mean, median and mode
- Population mean vs sample mean

$$\mu = \frac{1}{n} \sum_{i=1}^n a_i = \frac{1}{n} (a_1 + a_2 + \dots + a_{n-1} + a_n)$$

$$\hat{\mu} = \frac{1}{n} \sum_{i=1}^n r_i = \frac{1}{n} (r_1 + r_2 + \dots + r_{n-1} + r_n)$$

Averages

- Mean of a random variable

- ▣ Discrete

$$\mu = \sum_{i=1}^n p_i x_i$$

- ▣ Continuous

$$\mu = \int_{x_{\min}}^{x_{\max}} x f(x) dx$$

Expectations

- Example with one variable:

$$E[X] = 0.50 \cdot \$3 + 0.50 \cdot \$7 = \$5$$

- Rules for combining random variables and constants

$$E[X + Y] = E[X] + E[Y]$$

$$E[cX] = cE[X]$$

$$E[XY] \neq E[X]E[Y]$$

$$E[X^2] \neq E[X]^2$$

Variance & Standard Deviation

□ Definition

▣ Variance: $\sigma^2 = E[(X - \mu)^2]$

▣ Standard deviations: $\sigma = \sqrt{\sigma^2}$

□ Known PDF

$$\sigma^2 = \int_{x_{\min}}^{x_{\max}} (x - \mu)^2 f(x) dx$$

□ Sample estimate

$$\hat{\sigma}_x^2 = \frac{1}{n-1} \sum_{i=1}^n (x_i - \hat{\mu}_x)$$

Variance & Standard Deviation

$$\sigma^2 = E[X^2] - \mu^2 = E[X^2] - E[X]^2$$

□ Remember: $E[X^2] \neq E[X]^2$

Variance & Standard Deviation

- Rules for combining random variables and constants

$$\sigma[cX] = c\sigma[X]$$

$$\sigma[X + c] = \sigma[X]$$

Standardized Variables

The diagram illustrates the formula for a standardized variable, $Y = \frac{X - \mu}{\sigma}$. It includes four labels with arrows pointing to the corresponding parts of the formula: 'Standardized variable' points to Y , 'Random variable' points to X , 'Mean of X' points to μ , and 'Standard deviation of X' points to σ .

$$Y = \frac{X - \mu}{\sigma}$$

Standardized variable

Random variable

Mean of X

Standard deviation of X

Covariance & Correlation

□ Covariance

$$\sigma_{XY} = E[(X - \mu_X)(Y - \mu_Y)]$$

$$\hat{\sigma}_{X,Y} = \frac{1}{n-1} \sum_{i=1}^n (x_i - \hat{\mu}_X)(y_i - \hat{\mu}_Y)$$

□ Correlation

$$\rho_{XY} = \frac{\sigma_{XY}}{\sigma_X \sigma_Y}$$

Hedging

□ Portfolio variance

$$\sigma_{A+B}^2 = \sigma_A^2 + \sigma_B^2 + 2\rho_{AB}\sigma_A\sigma_B$$

□ Optimal hedging $\Rightarrow$ minimize variance

$$P = X_A + hX_B$$
$$\sigma_P^2 = \sigma_A^2 + h^2\sigma_B^2 + 2h\rho_{AB}\sigma_A\sigma_B$$

$$h^* = -\rho_{AB} \frac{\sigma_A}{\sigma_B}$$

Moments

□ Mean:

$$\mu = E[X]$$

□ k th moment:

$$m_k = E[X^k]$$

□ Central moment:

$$\mu_k = E[(X - \mu)^k]$$

Skewness & Kurtosis

□ Skewness

$$\text{Skewness} = \frac{E[(X - \mu)^3]}{\sigma^3}$$

$$\tilde{s} = \frac{n}{(n-1)(n-2)} \sum_{i=1}^n \left(\frac{x_i - \hat{\mu}}{\hat{\sigma}} \right)^3$$

□ Kurtosis

$$K = \frac{E[(X - \mu)^4]}{\sigma^4}$$

$$K_{\text{excess}} = K - 3$$

$$\tilde{K} = \frac{n(n+1)}{(n-1)(n-2)(n-3)} \sum_{i=1}^n \left(\frac{x_i - \hat{\mu}}{\hat{\sigma}} \right)^4$$

$$\tilde{K}_{\text{excess}} = \tilde{K} - 3 \frac{(n-1)^2}{(n-2)(n-3)}$$

Coskewness & Cokurtosis

□ Two portfolios:

	1	2	3	4	5		mean	std. dev.	correl	skew
A	-20%	-10%	0%	10%	20%		0%	16%		0
B	-20%	0%	20%	10%	-10%		0%	16%	30%	0
(A+B)/2	-20%	-5%	10%	10%	5%		0%	13%		-1.21
	1	2	3	4	5		mean	std. dev.	correl	skew
C	20%	10%	0%	-10%	-20%		0%	16%		0
D	20%	0%	-20%	-10%	10%		0%	16%	30%	0
(C+D)/2	20%	5%	-10%	-10%	-5%		0%	13%		1.21

Coskewness & Cokurtosis

□ Coskewness

$$S_{AAB} = E[(A - \mu_A)^2 (B - \mu_B)] / \sigma_A^2 \sigma_B$$

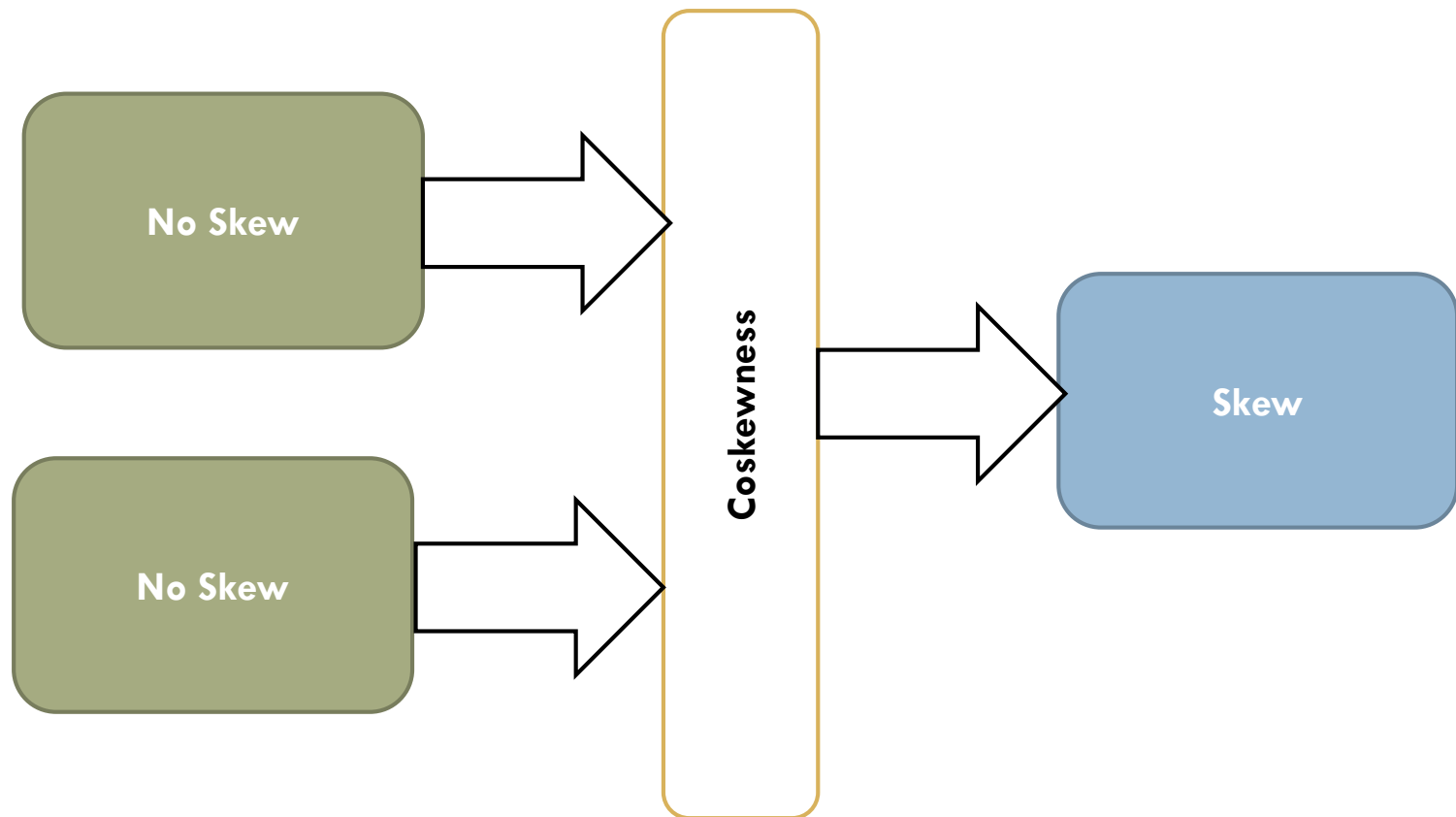
$$S_{ABB} = E[(A - \mu_A) (B - \mu_B)^2] / \sigma_A \sigma_B^2$$

□ Cokurtosis

$$K_{AAAB} = \frac{E[(A - \mu_A)^3 (B - \mu_B)]}{\sigma_A^3 \sigma_B}$$

⋮

Coskewness & Cokurtosis



Coskewness & Cokurtosis

- Pros

- ▣ Similar to variance, covariance, skewness, and kurtosis

- Cons

- ▣ Very large number of cross moments
 - ▣ Not widely used in other fields

BLUE

- Best Linear Unbiased Estimator

Mathematics & Statistics for Financial Risk Management

4: Distributions

Topics

- Parametric Distribution
- Uniform Distribution
- Bernoulli & Binomial Distributions
- Poisson Distribution
- Normal Distribution
- Lognormal Distribution
- Central Limit Theorem
- Monte Carlo Simulations
- Chi-Squared Distribution
- Student's t Distribution
- F Distribution
- Triangular and Beta Distributions
- Mixture Distributions

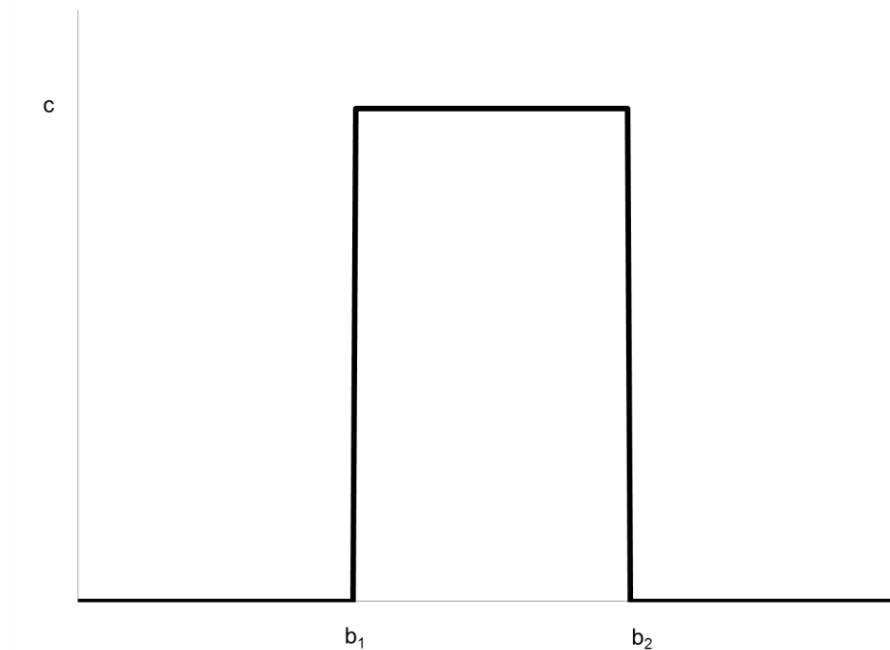
Parametric Distributions

- What is a parametric distribution?

Uniform Distribution

□ PDF

$$u(b_1, b_2) = \begin{cases} c & \forall b_1 \leq x \leq b_2 \\ 0 & \forall b_1 > x > b_2 \end{cases} \quad s.t. \ b_2 > b_1 \quad c = \frac{1}{b_2 - b_1}$$



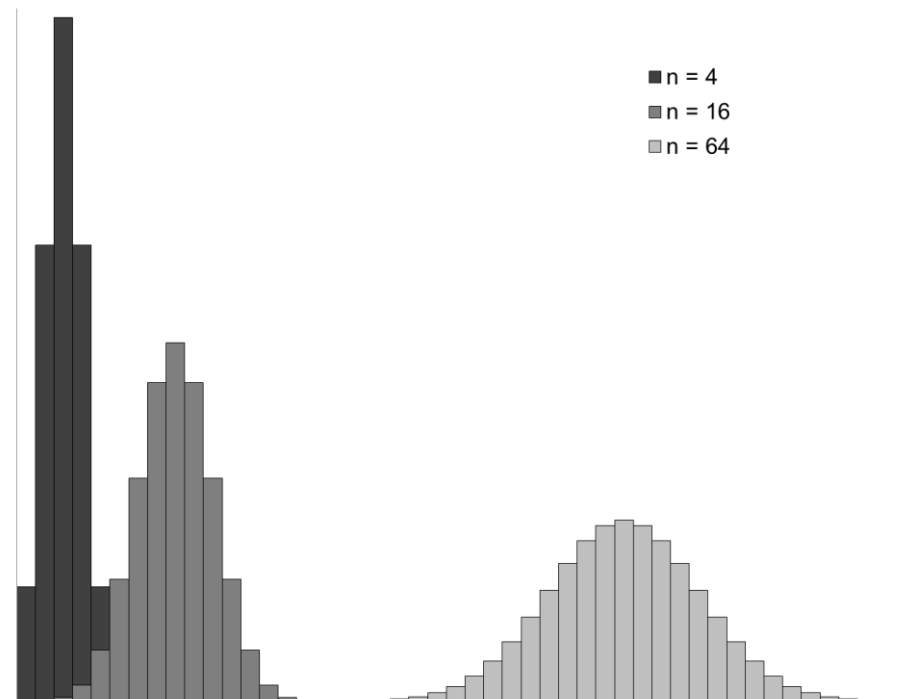
Bernoulli & Binomial Distribution

□ Bernoulli Trial

$$P[X = 1] = p \text{ and } P[X = 0] = 1 - p$$

□ Binomial Distribution

$$P[K = k] = \binom{n}{k} p^k (1 - p)^{n-k}$$



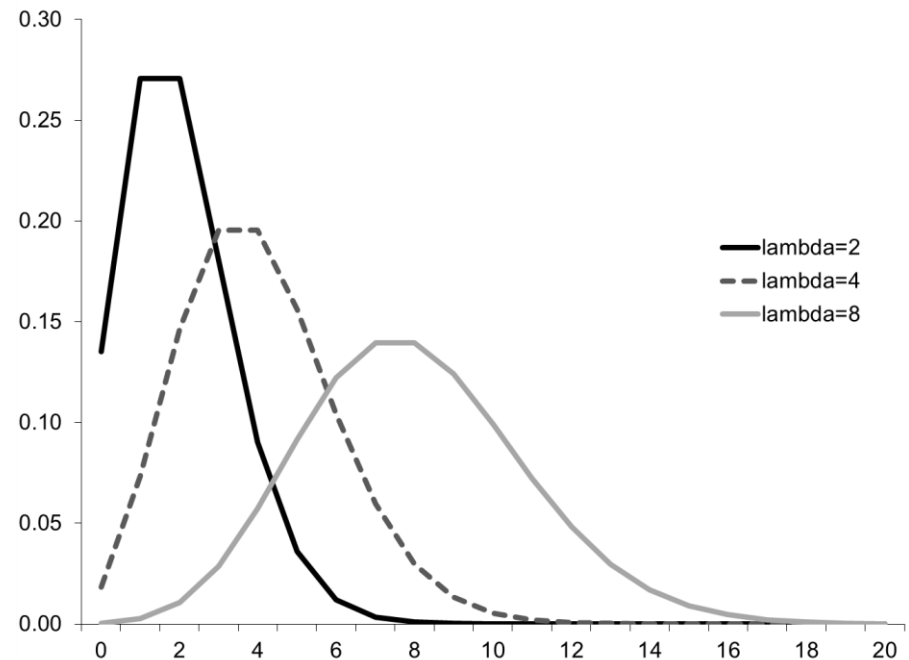
Poisson Distribution

□ Poisson random variable:

$$P[X = n] = \frac{\lambda^n}{n!} e^{-\lambda}$$

□ Poisson process

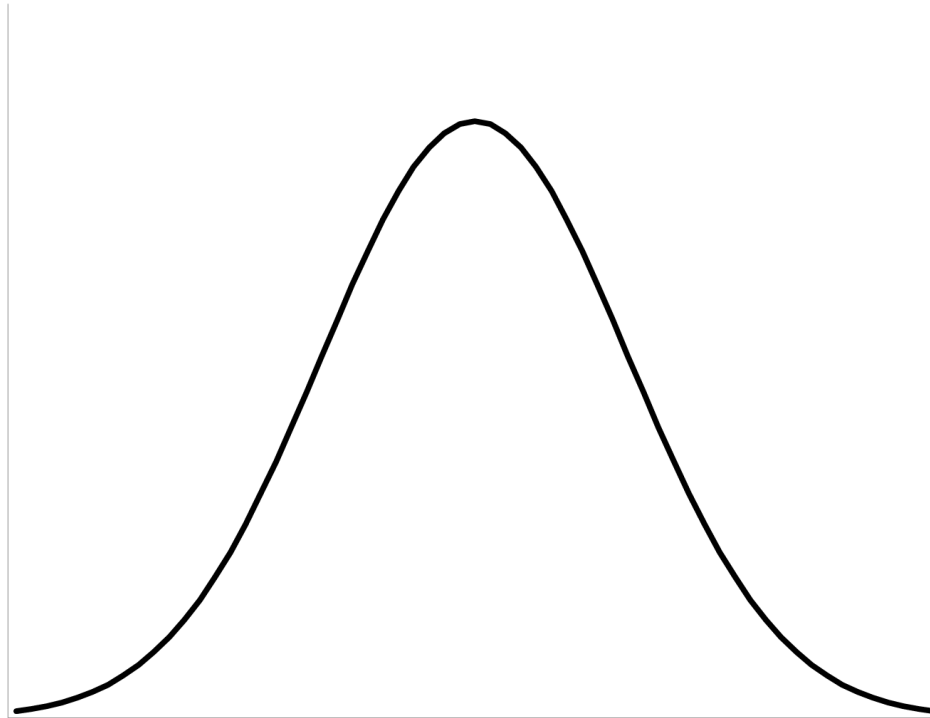
$$P[X = n] = \frac{(\lambda t)^n}{n!} e^{-\lambda t}$$



Normal Distribution

□ PDF:

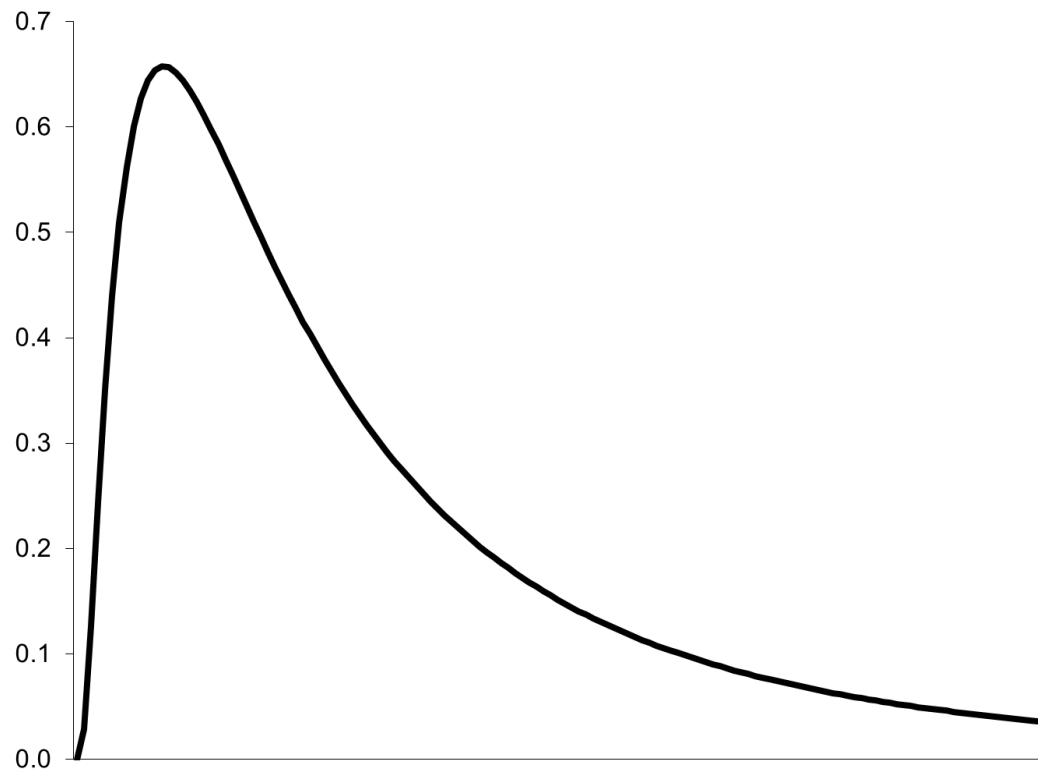
$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}$$



Lognormal Distribution

□ PDF

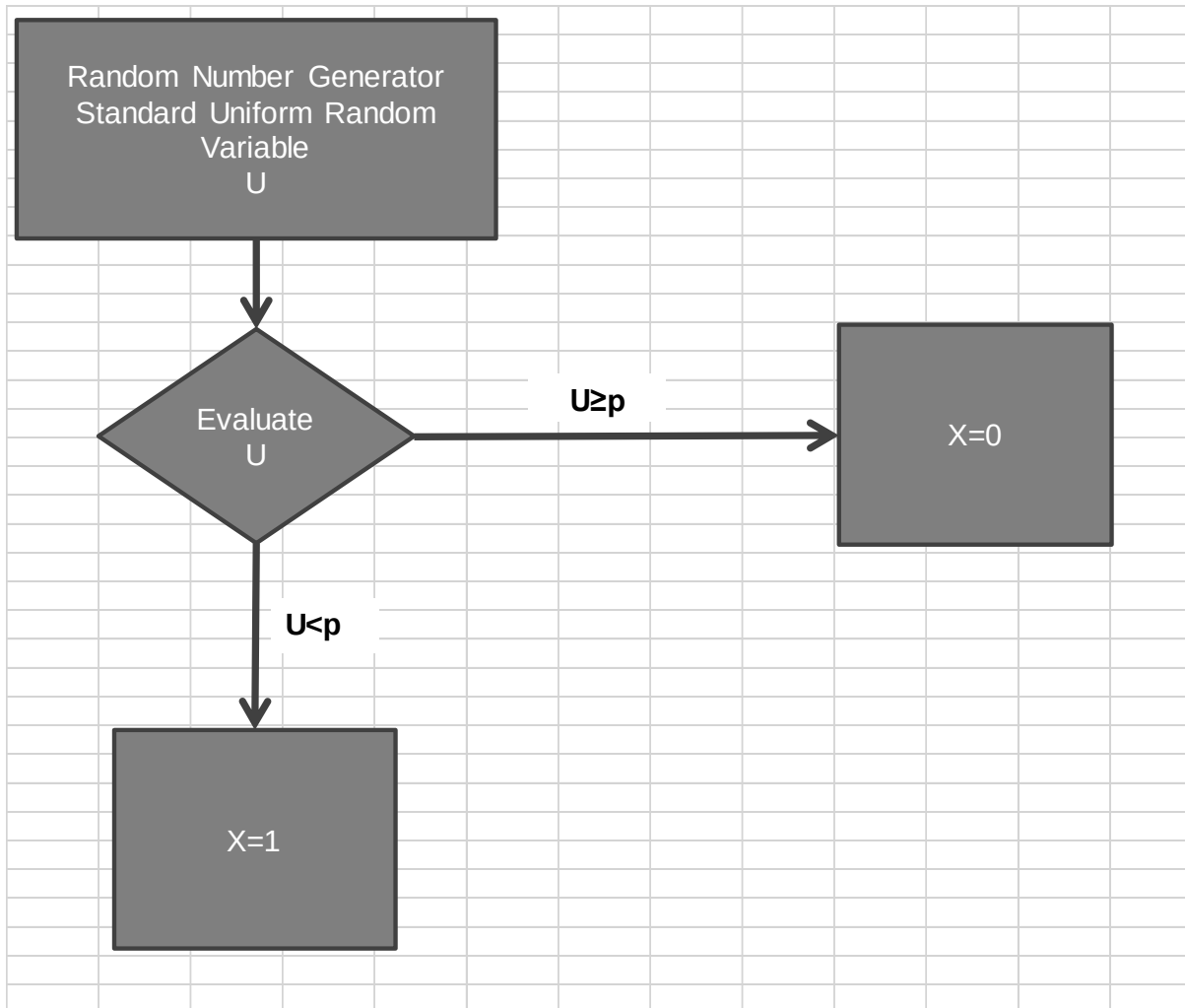
$$f(x) = \frac{1}{x\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{\ln x - \mu}{\sigma}\right)^2}$$



Central Limit Theorem

$$\lim_{n \rightarrow \infty} S_n \sim N(n\mu, n\sigma^2)$$

Montel Carlo Simulation

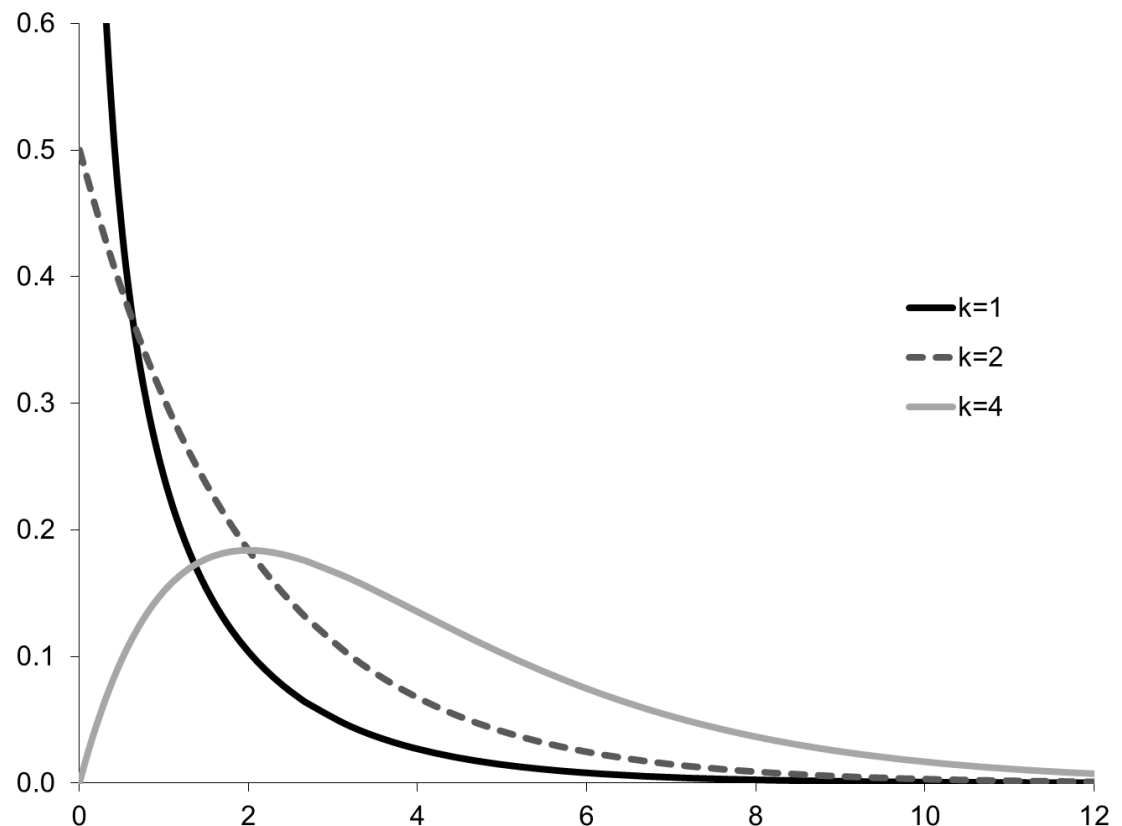


Chi-Squared Distribution

- k independent standard normal variables, $Z_1, Z_2, \dots, Z_k$

$$S = \sum_{i=1}^k Z_i^2$$

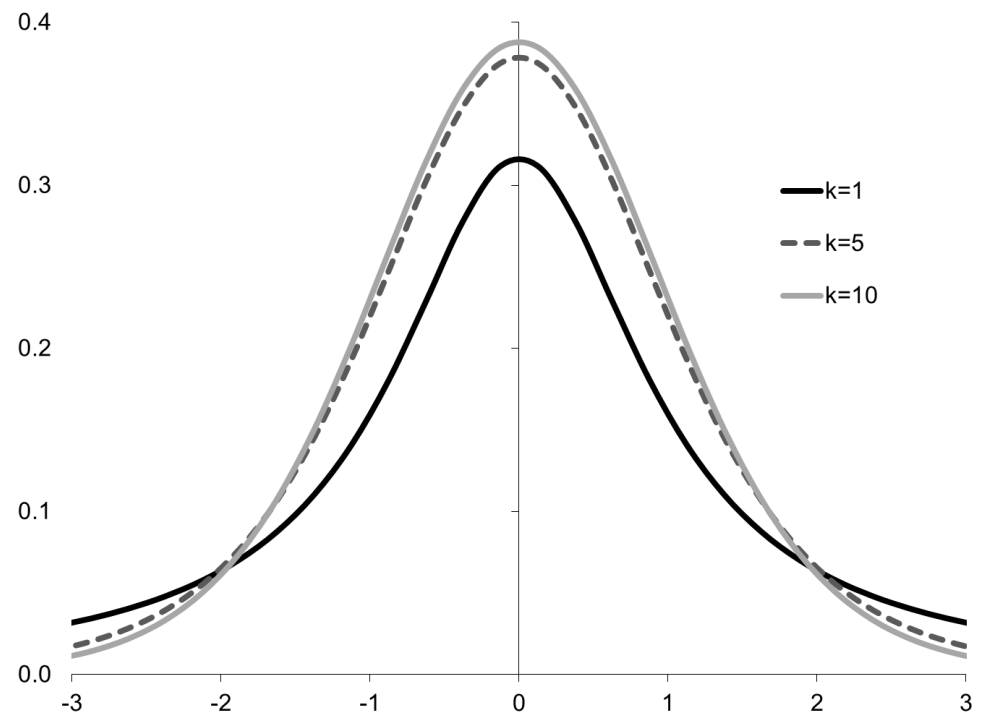
$$S \sim \chi_k^2$$



Student's t Distribution

$$X = \frac{Z}{\sqrt{U/k}}$$

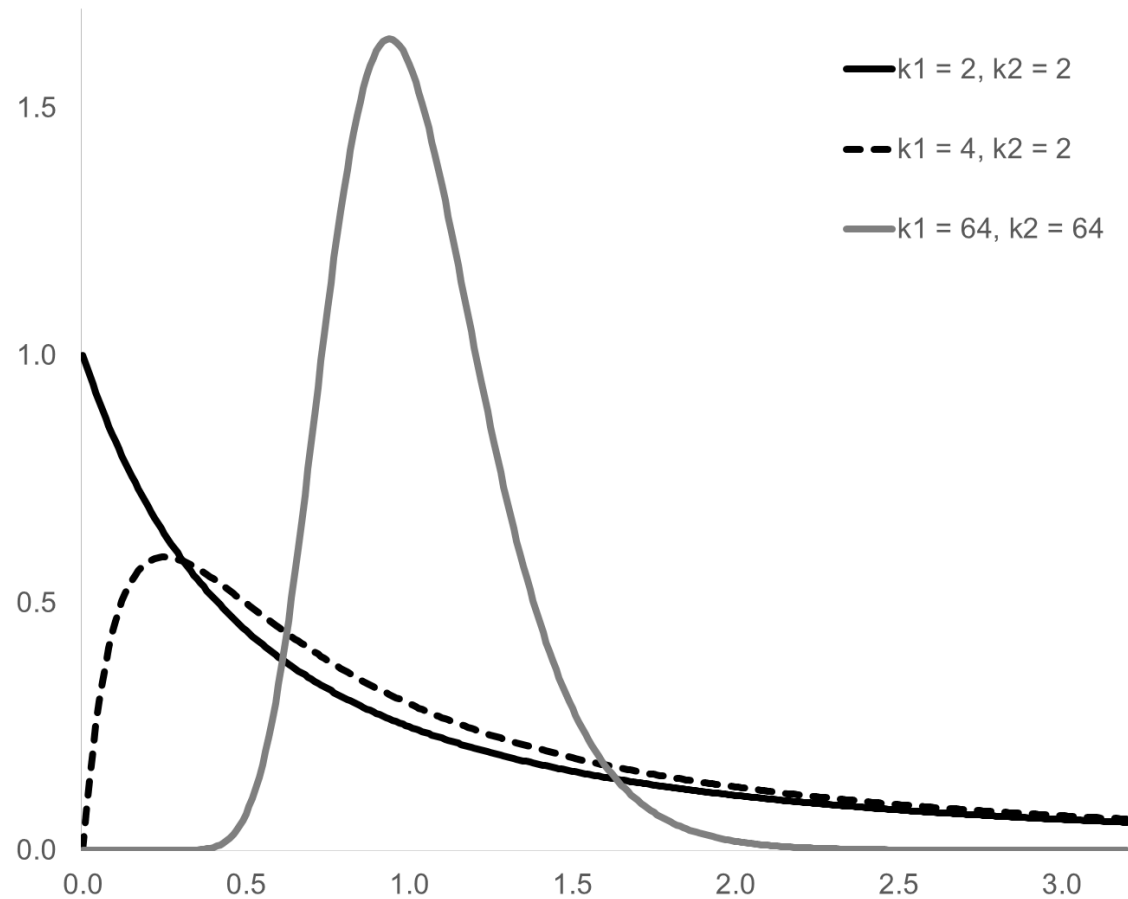
$$f(x) = \frac{\Gamma\left(\frac{k+1}{2}\right)}{\sqrt{k\pi}\Gamma\left(\frac{k}{2}\right)} \left(1 + \frac{x^2}{k}\right)^{-(k+1)/2}$$



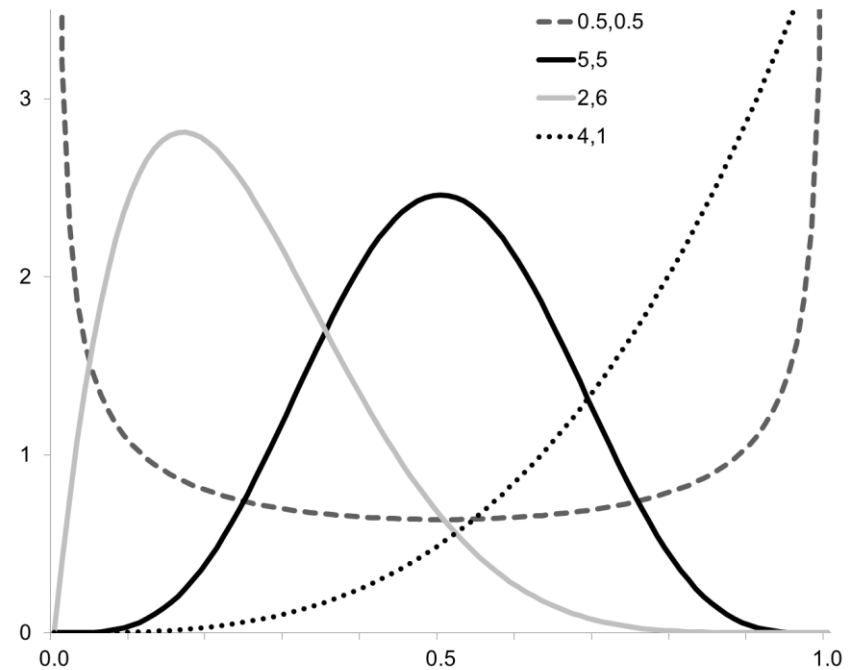
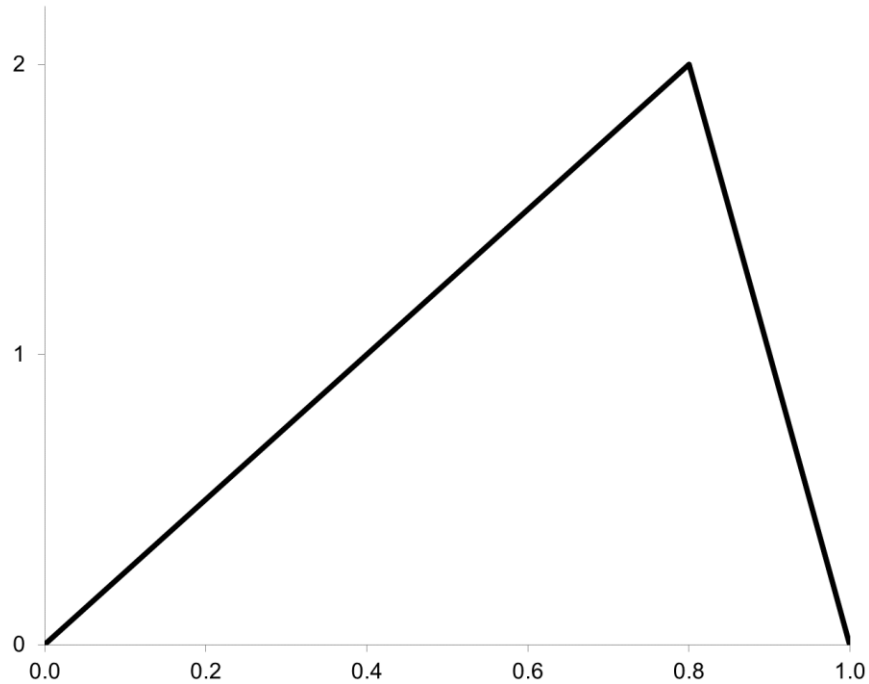
F Distribution

$$X = \frac{U_1 / k_1}{U_2 / k_2} \sim F(k_1, k_2)$$

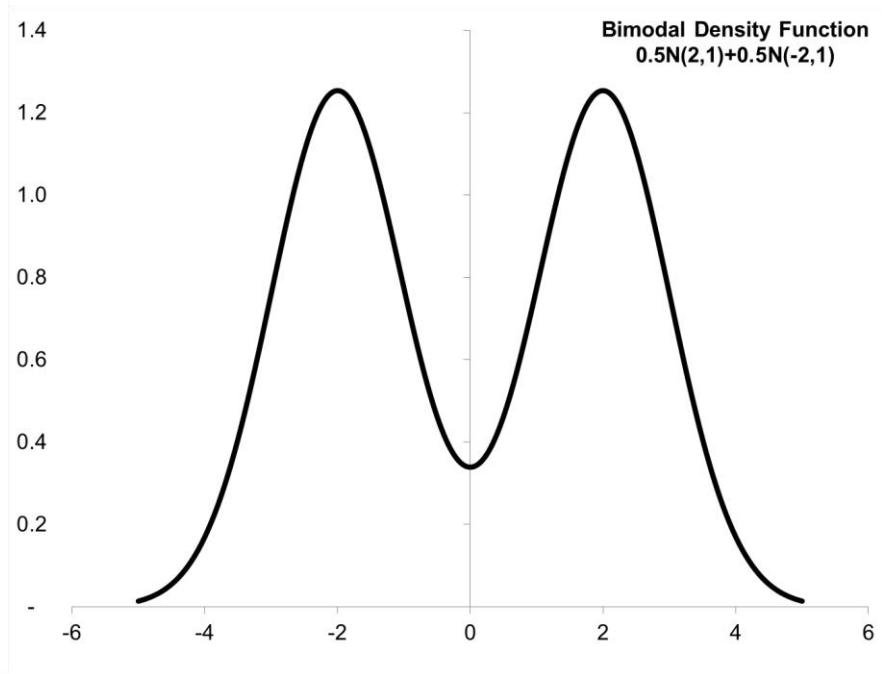
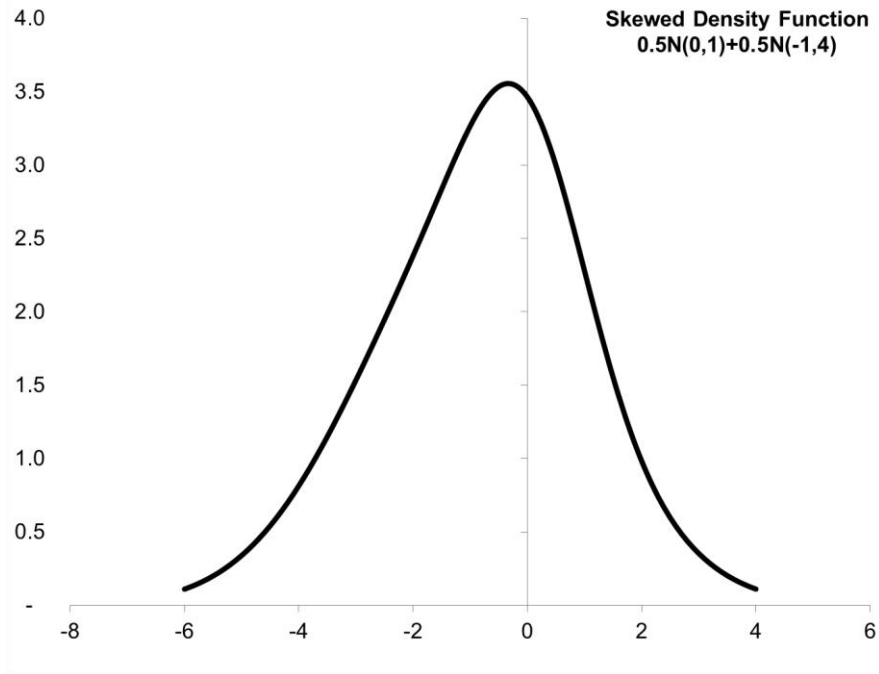
$$f(x) = \frac{\sqrt{\frac{(k_1 x)^{k_1} k_2^{k_2}}{(k_1 x + k_2)^{k_1 + k_2}}}}{xB\left(\frac{k_1}{2}, \frac{k_2}{2}\right)}$$



Triangular & Beta Distributions



Mixture Distributions



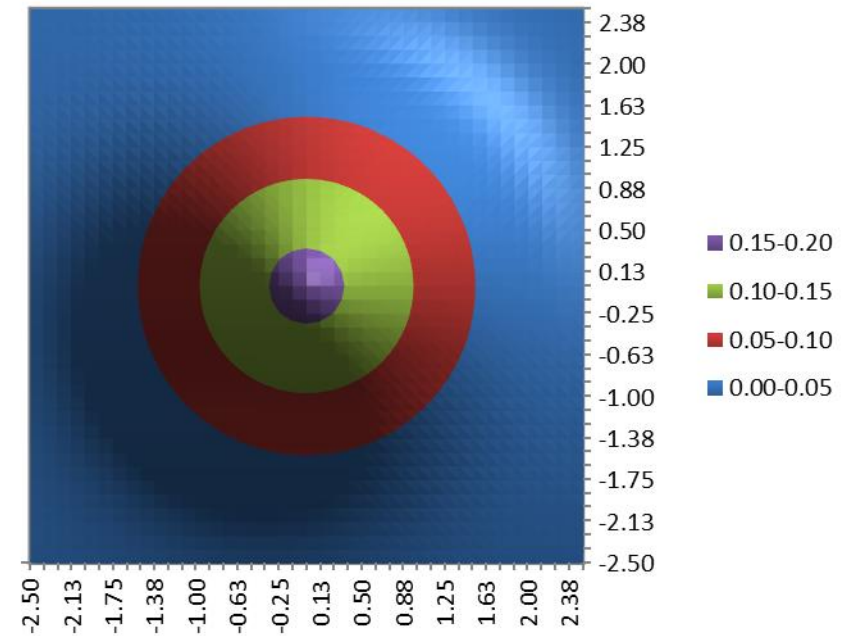
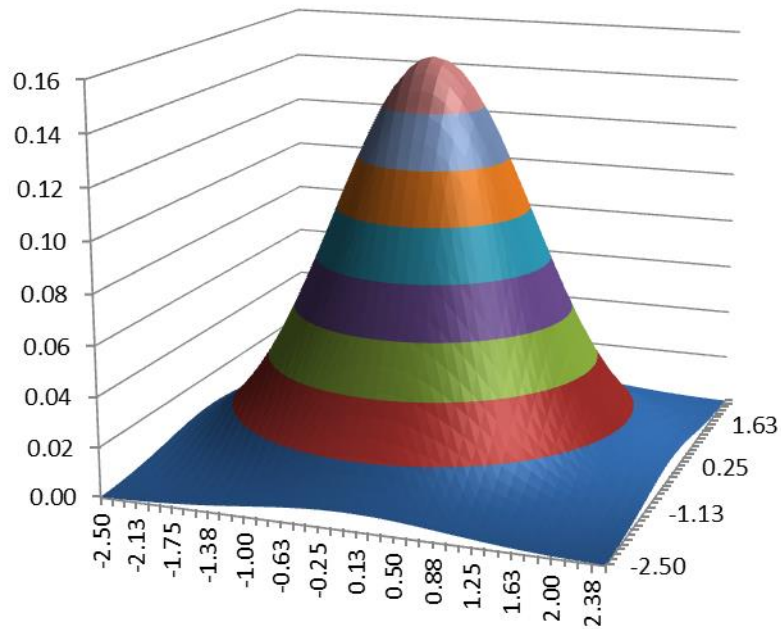
Mathematics & Statistics for Financial Risk Management

5: Multivariate Distribution & Copulas

Topics

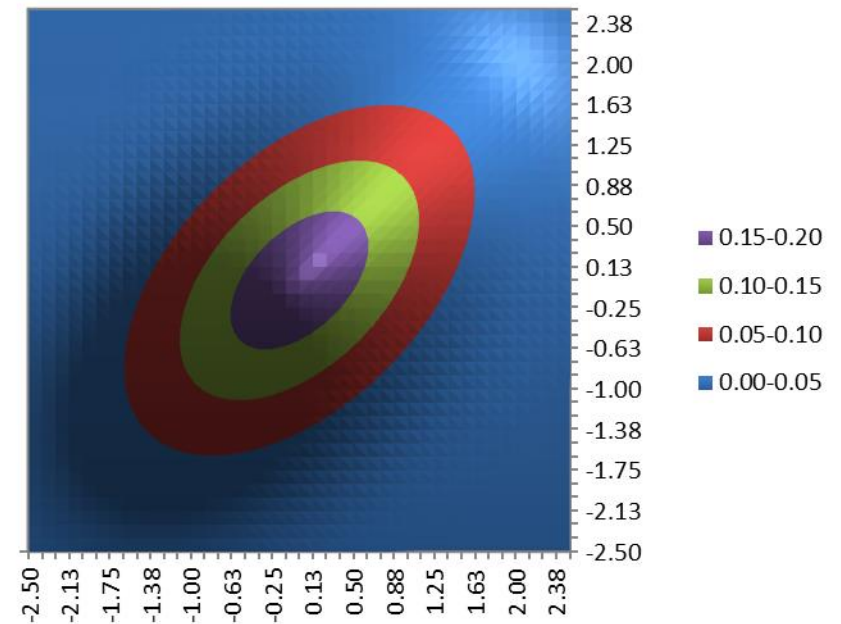
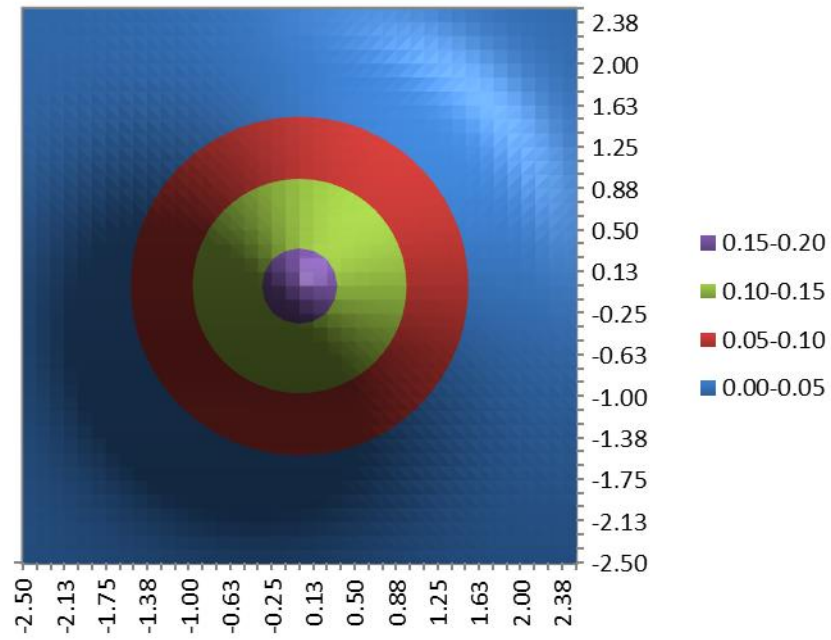
- Multivariate Distributions
- Copulas
- Coskewness Revisited

Multivariate Distributions



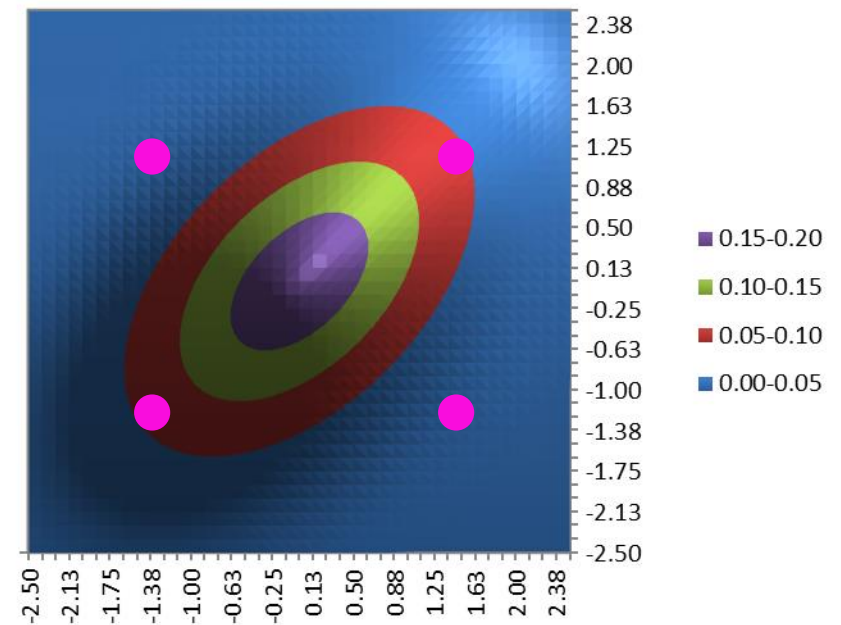
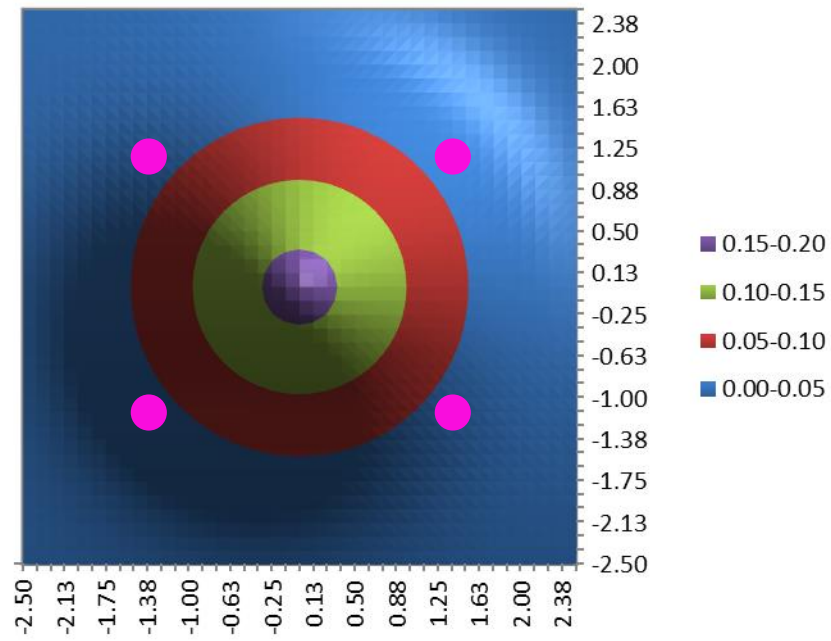
Multivariate Distributions

□ No Correlation vs Correlation



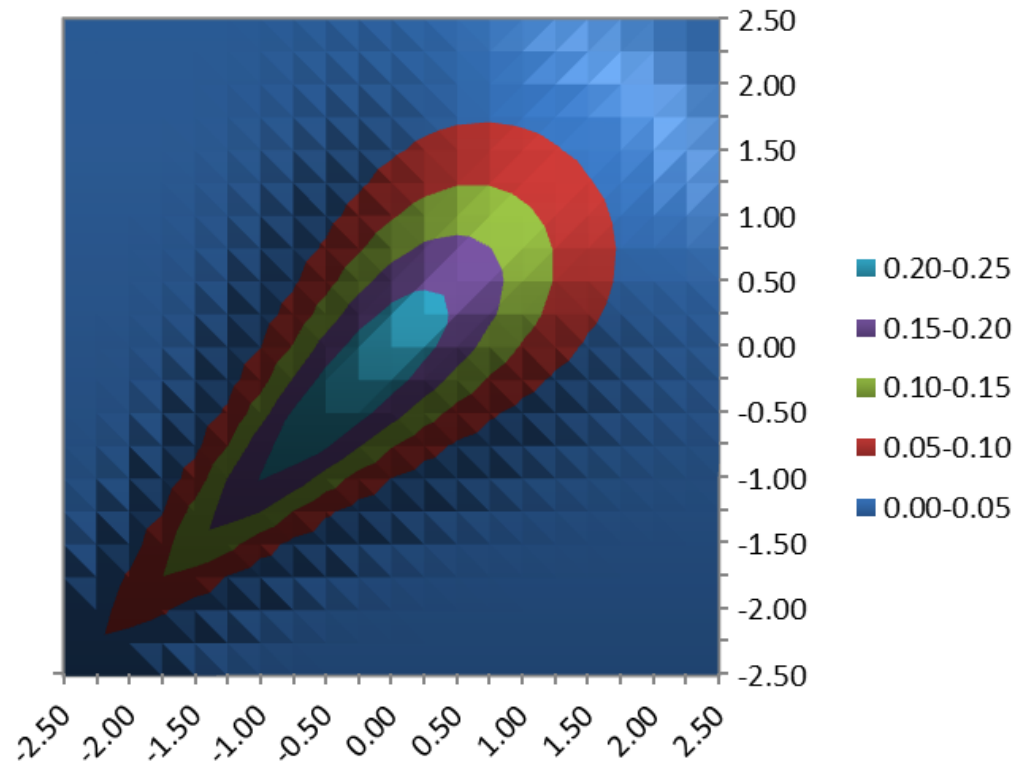
Multivariate Distributions

□ No Correlation vs Correlation

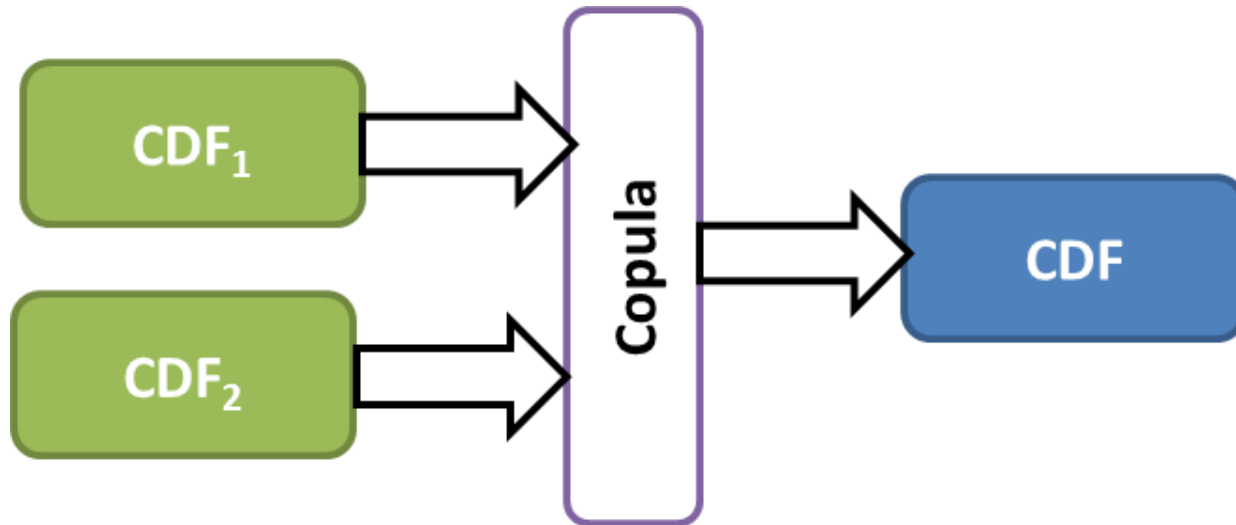


Copulas

□ Clayton Copula

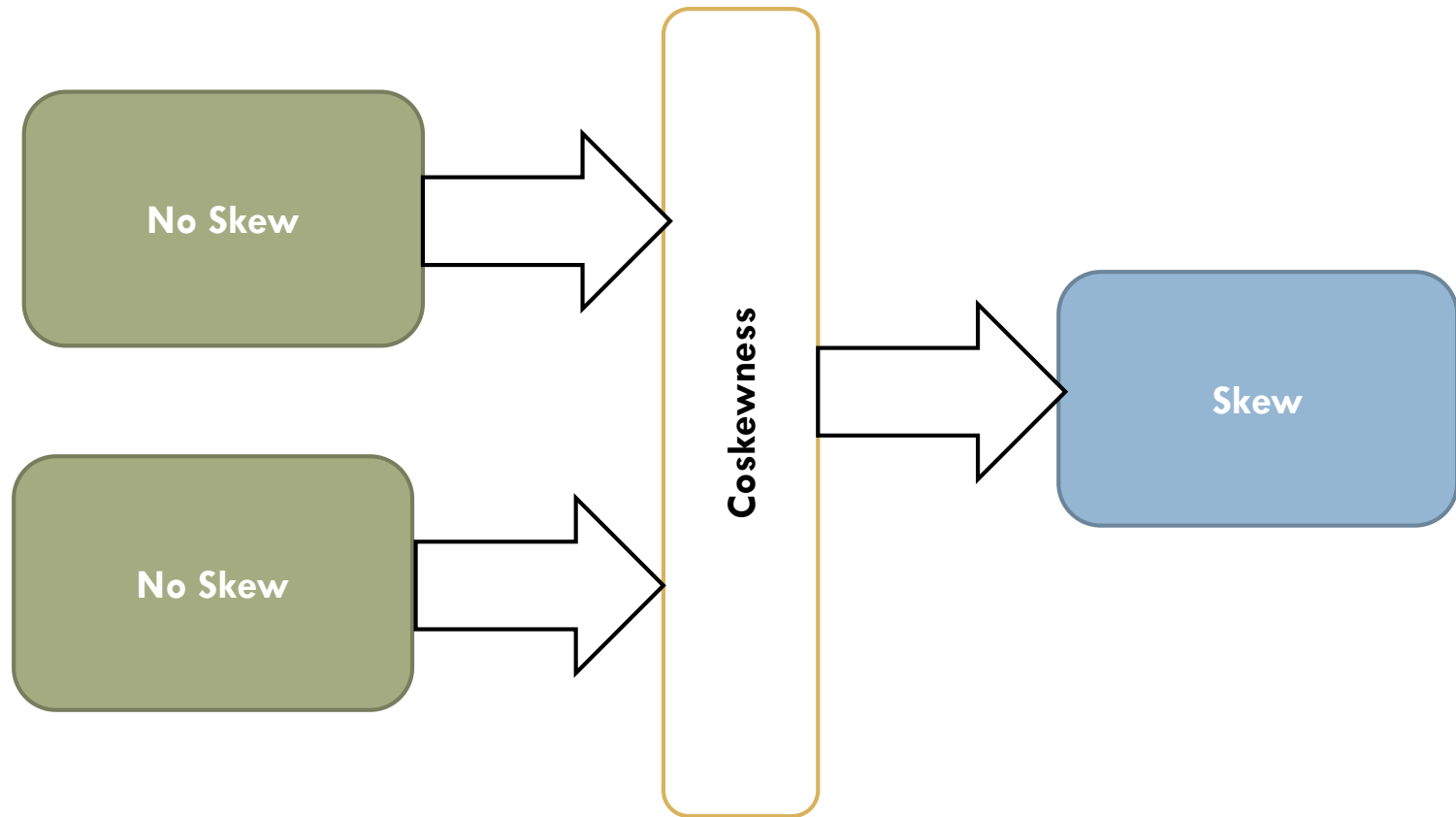


Copulas

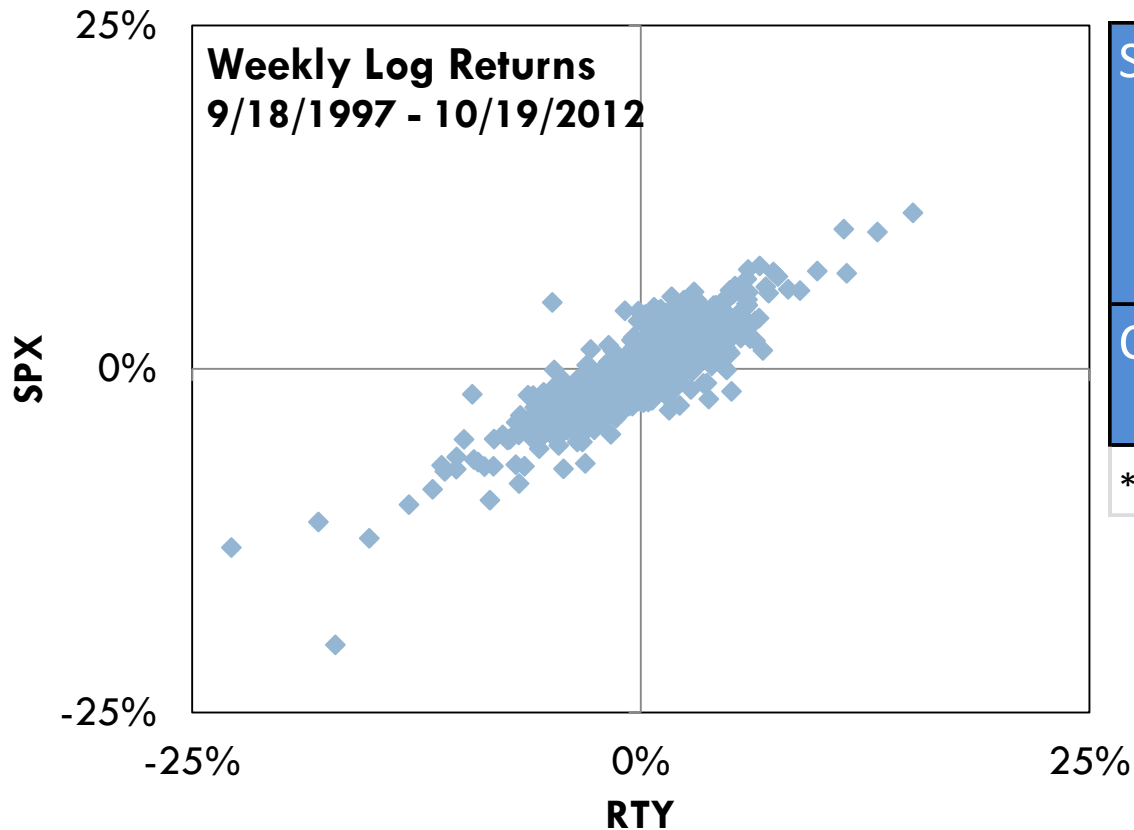


$$\text{Franks Copula: } C(u, v) = \frac{1}{\alpha} \ln \left[1 + \frac{(e^{\alpha u} - 1)(e^{\alpha v} - 1)}{e^{\alpha} - 1} \right]$$

Coskewness



Coskewness



Skew	RTY	-1.03
	SPX	-0.84
	RTY+SPX	-1.01
	RTY+SPX *	-0.28
Coskew	RRS	-0.90
	RSS	-0.84

*skew if no coskew

Copulas

- Pros

- ▣ Very flexible
- ▣ Used in other fields

- Cons

- ▣ Math is complicated
- ▣ Not intuitive?

Mathematics & Statistics for Financial Risk Management

6: Bayesian Analysis

Topics

- Bayesian Analysis
 - ▣ History
 - ▣ Bayes' Theorem
 - ▣ Bayesian Networks
 - ▣ Bayesians vs Frequentists

Bayes' Theorem

□ History

- ▣ Thomas Bayes (1701-1761)
- ▣ Pierre-Simon Laplace publishes in 1812
- ▣ WW II (1939-1945) code breaking

□ Modern Applications

- ▣ Medicine
- ▣ AI: spam filters, machine translation, machine vision
- ▣ Risk management?

Bayes in the news

In HP-Autonomy debacle, many advisers but little good advice



61 people recommend this.

By Nadia Damouni and Nicola Leske

Wed Nov 21, 2012 5:15am EST

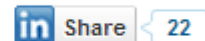
(Reuters) - When Hewlett Packard acquired Autonomy last year for \$11.1 billion, some 15 different financial, legal and accounting firms were involved in the transaction -- and none raised a flag about what HP said Tuesday was a major accounting fraud.

HP stunned Wall Street with the allegations about its British software unit and took an \$8.8 billion writedown, the latest in a string of reversals for the storied company.

HP Chief Executive Meg Whitman, who was a director at the company at



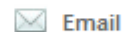
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22



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Related News

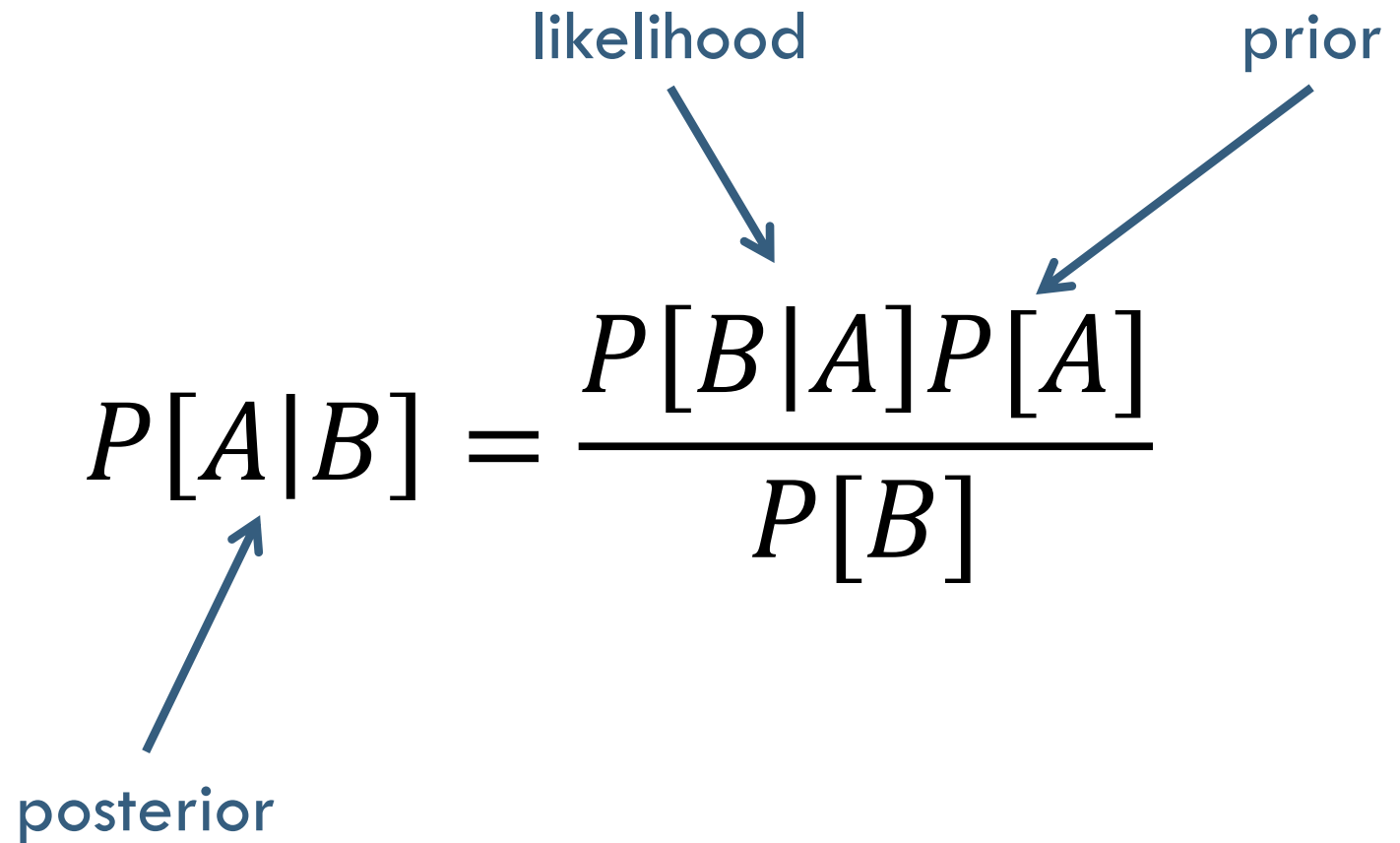
[HP accuses Autonomy of wrongdoing, takes \\$8.8 billion charge](#)

Wed, Nov 21 2012

Bayes' Theorem

$$P[A|B] = \frac{P[B|A]P[A]}{P[B]}$$

Bayes' Theorem



A diagram illustrating Bayes' Theorem. The equation is $P[A|B] = \frac{P[B|A]P[A]}{P[B]}$. Three blue arrows point to parts of the equation: one from the word 'likelihood' to $P[B|A]$, one from the word 'prior' to $P[A]$, and one from the word 'posterior' to $P[A|B]$.

likelihood

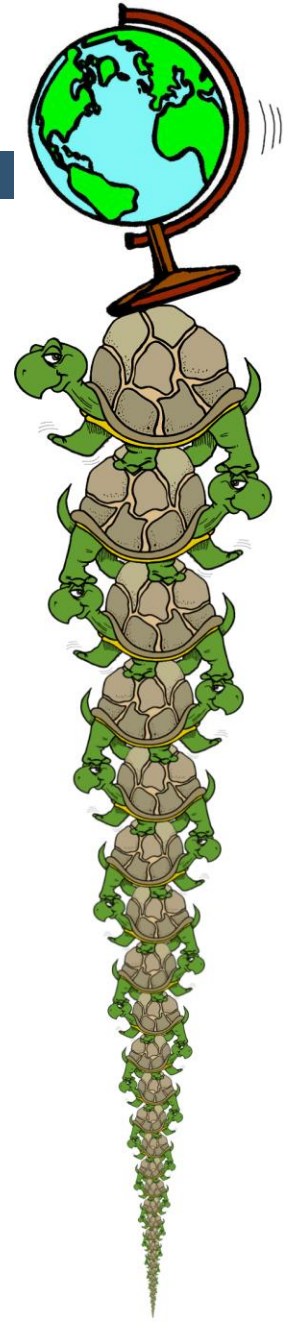
prior

$$P[A|B] = \frac{P[B|A]P[A]}{P[B]}$$

posterior

Bayesian Analysis

- Where do prior assumptions come from?



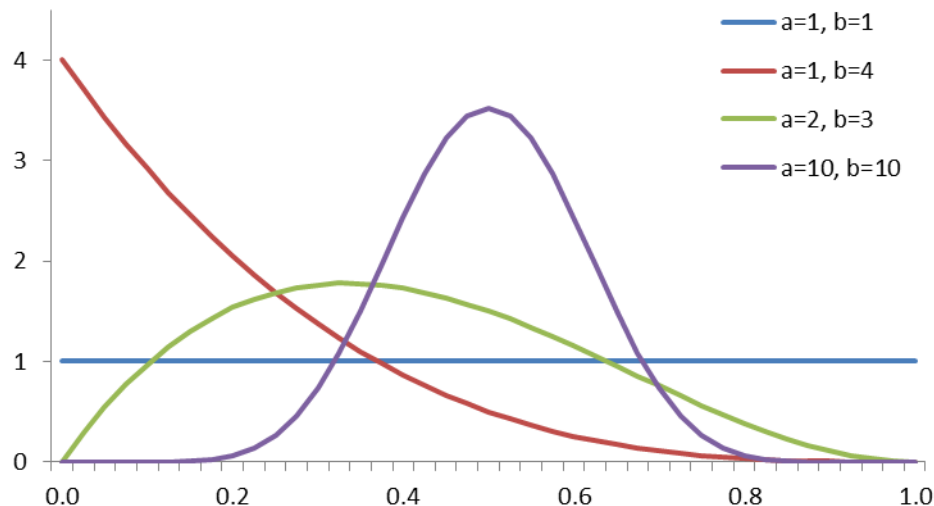
Simple to Complex

□ Coin flipping revisited

▣ Instead of 2 possible probabilities (biased or unbiased), what if there where 10? 100? Infinitely many?

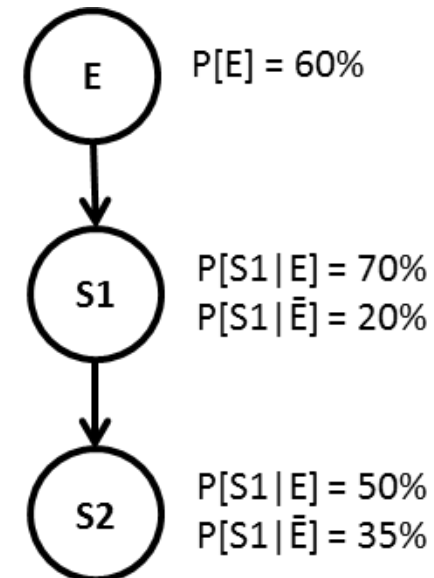
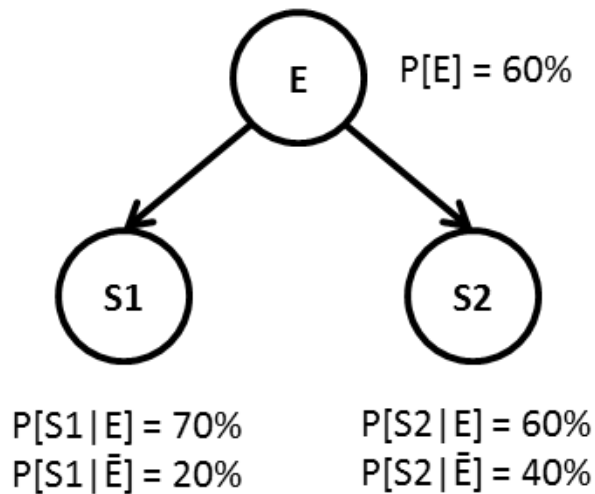
▣ Beta distribution:

$$f(x) = \frac{x^{\alpha-1}(1-x)^{\beta-1}}{B(\alpha, \beta)}$$



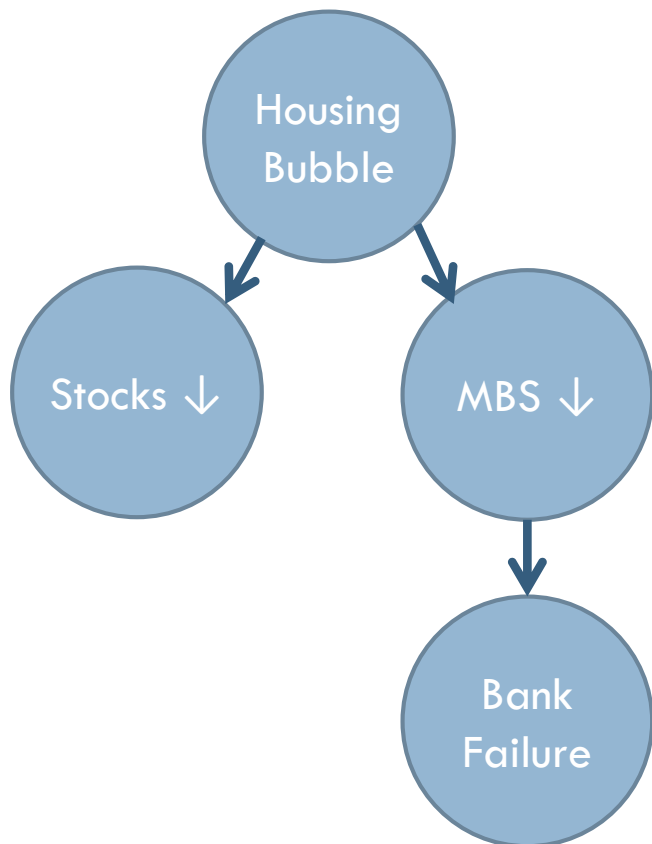
Bayesian Networks

- Causality is implicit
- Fewer parameters



Bayesian Network

- Easier to identify experts
- Conditional probability is often more intuitive



	H	S	M	B
H	0.40	0.10	0.33	0.49
S	0.10	0.36	0.27	0.04
M	0.33	0.27	0.08	0.21
B	0.49	0.04	0.21	0.55

Bayesian Analysis

Frequentists vs Bayesians



Further Reading

- *Introduction to Bayesian Statistics*. William Bolstad.
- *Coherent Stress Testing: A Bayesian Approach to the Analysis of Financial Stress*. Riccardo Rebonato.
- *The Theory That Would Change the World*. Sharon Betsch McGrayne.
- *The Signal and the Noise*. Nate Silver.

Mathematics & Statistics for Financial Risk Management

7: Hypothesis Testing & Confidence Intervals

Topics

- Sample Mean Revisited
- Sample Variance Revisited
- Confidence Intervals
- Hypothesis Testing
- Chebyshev's Inequality
- VaR

Sample Mean

- The sample mean itself is a random variable

$$E[\hat{\mu}] = \mu$$

$$\sigma_{\hat{\mu}}^2 = \frac{\sigma^2}{n}$$

Sample Variance

- The sample variance is also a random variable

$$E[(\hat{\sigma}^2 - \sigma^2)^2] = \sigma^4 \left(\frac{2}{n-1} + \frac{\kappa}{n} \right)$$

$$(n-1) \frac{\hat{\sigma}^2}{\sigma^2} \sim \chi_{n-1}^2$$

Confidence Interval

- Standardized version of our estimator has a t distribution.

$$t = \frac{\hat{\mu} - \mu}{\hat{\sigma} / \sqrt{n}}$$

Hypothesis Testing

- Example: $H_0 : \mu_r > 10\%$

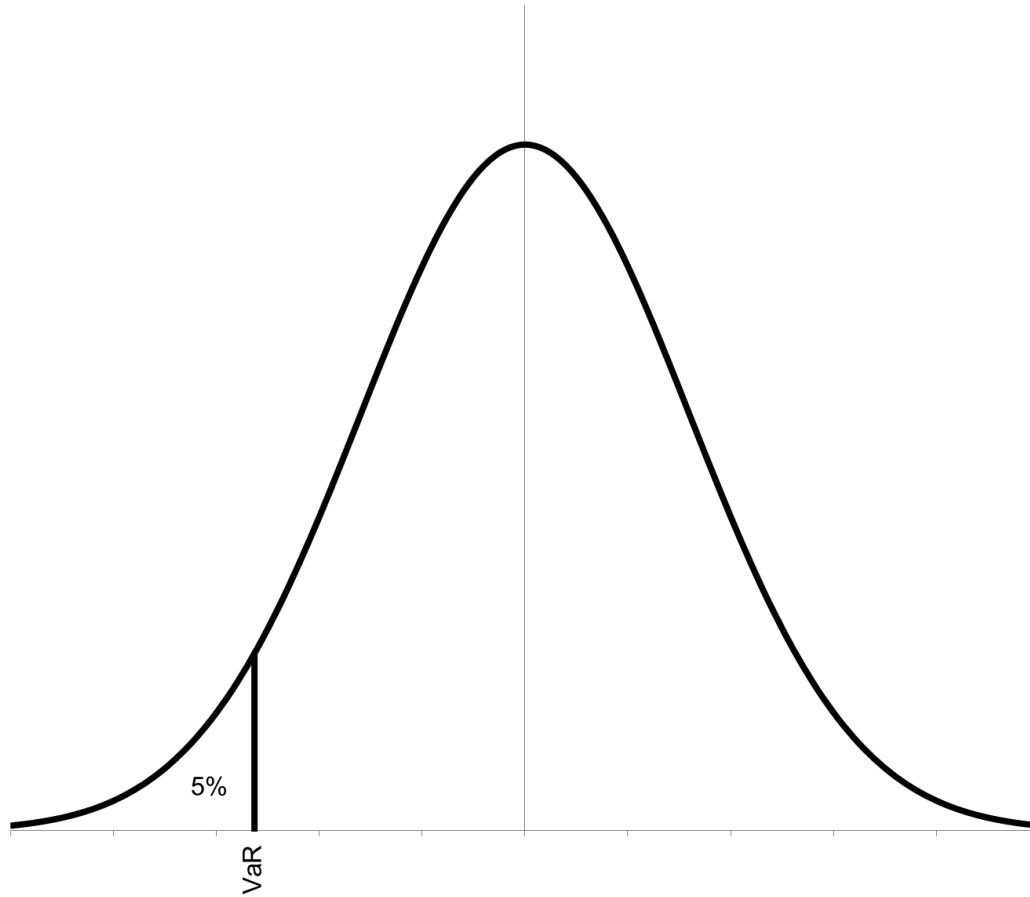
Chebyshev's Inequality

- Even when distribution is unknown, we know:

$$P[|X - \mu| \geq n\sigma] \leq \frac{1}{n^2}$$

VaR

- Value at risk



VaR

□ Pros

- ▣ One number
- ▣ Focus on losses
- ▣ Easy to understand
- ▣ Widely used

□ Cons?

- ▣ Model complexity
- ▣ Shape of tail?
- ▣ Subadditivity?
- ▣ Doesn't work?

Backtesting

- Is 99% better than 95%?
- Use binomial distribution to test
 - ▣ Probability of an exceedance
 - ▣ Probability given exceedance (day after subset)
 - ▣ Probability given risk level (subsets by risk level)

$$P[K = k] = \binom{n}{k} p^k (1 - p)^{n-k}$$

- Information in model failure?

VaR

Actual Exceedances 2007

	α	Actual	$E[X]$	$P[x \leq A]$	$P[x \geq A]$
Bank of America	99%	14	2.6	100.0%	0.0%
Credit Suisse	99%	15	2.6	100.0%	0.0%
JP Morgan	99%	8	2.6	99.9%	0.5%
UBS	99%	29	2.6	100.0%	0.0%
Wachovia	99%	27	2.6	100.0%	0.0%
Bear Stearns	95%	27	12.8	100.0%	0.0%
Goldman Sachs	95%	10	12.8	26.3%	82.7%
Lehman Brothers	95%	4	12.8	0.4%	99.9%
Morgan Stanley	95%	15	12.8	78.6%	30.2%

Source: Contreras, Patricio. "Is VAR a useful tool in volatile markets?". Risk Magazine, Oct 2010.

Getting better?

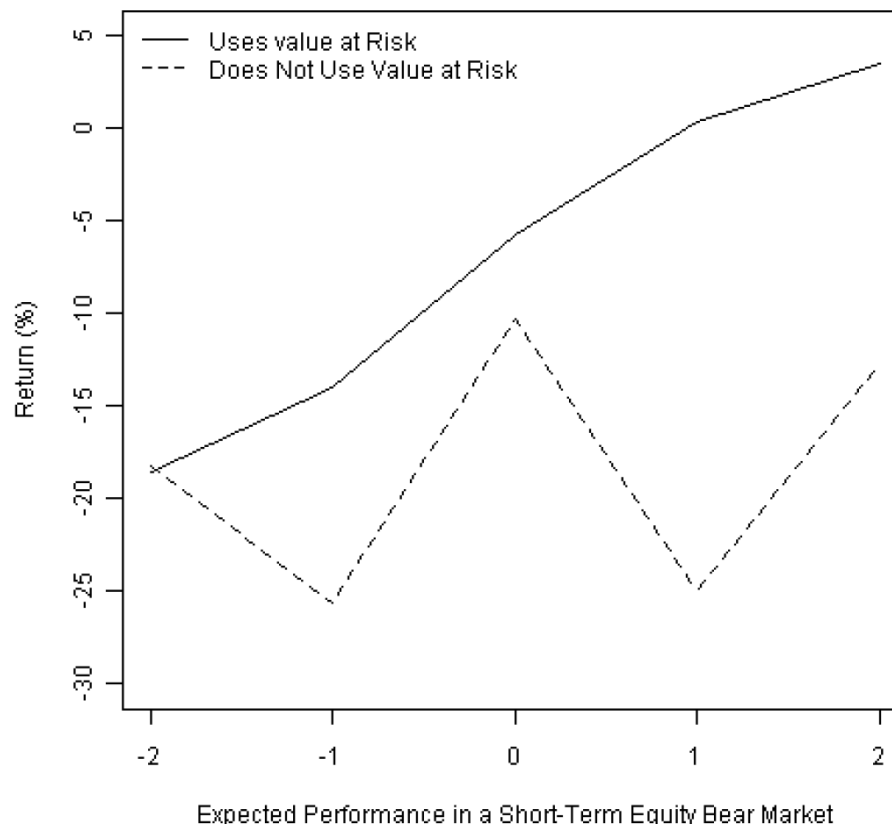
“...(VAR is) estimated, at a 99% confidence level ... Citi would thus expect the number of days where trading losses exceed its VAR to be less than two or three exceptions per year. Periods of unstable market conditions could increase the number of these exceptions. During the last four quarters, there was one back-testing exception where trading losses exceeded the VAR estimate at the Citigroup level” —Citi 10-K, 12/31/2011

“... Assuming no intra-day trading, for a 95%/one-day VaR, the expected number of times that trading losses should exceed VaR during the year is 13, and, in general, if trading losses were to exceed VaR more than 21 times in a year, the adequacy of the VaR model could be questioned.” —Morgan Stanley 10-K, 12/31/2011

“Our VaR model uses a historical simulation ... a 99 percent confidence level. ... Actual losses did not exceed daily trading VaR in 2011 or 2010.” —Bank of America 10-K, 12/31/2011

Hedge Fund VaR Use

Panel B: Median Performance



Source: Cassar, Gavin and Joseph Gerakos. "How do hedge funds manage portfolio risk?". March, 2011.

79 funds; Sep-Nov 2008 Performance vs expected performance in bear market prior to 2008 (-2 = down, -1 = down a little, 0 = no effect, 1 = up a little, 2 = up)

VaR

Vendors of risk software

- Bara
 - ▣ US equity –centric
 - ▣ Factor model
 - ▣ Purchased by MSCI
- RiskMetrics
 - ▣ JP Morgan spin-out
 - ▣ Popularized VaR
 - ▣ Statistics/Econometrics –centric
 - ▣ Purchased by MSCI
- Others

Mathematics & Statistics for Financial Risk Management

Topics

- Matrix Notation
- Matrix Operations
- Transition Matrices
- Monte Carlo & Cholesky Decomposition

Matrix Notation

□ Matrix:

$$\mathbf{A} = \begin{bmatrix} 3 & 5 \\ -9 & 3 \\ 10 & 8 \end{bmatrix}$$

□ Vector:

$$\mathbf{b} = \begin{bmatrix} 43 \\ 17 \\ -56 \\ 64 \end{bmatrix}$$

Matrix Operations

- Addition
- Subtraction
- Multiplication
- Transpose
- Inverse

Transition Matrices

5-Year		To a rating of:			
From a rating of:	A	A	B	C	D
	B	64.7%	24.8%	6.7%	3.7%
	C	28.1%	46.0%	12.1%	13.8%
	D	11.8%	35.0%	14.3%	39.0%
		0.0%	0.0%	0.0%	100.0%

$$\mathbf{T}_n = \mathbf{T}_1^n$$

Monte Carlo

□ Cholesky Decomposition:

Covariance matrix $\rightarrow \Sigma = \mathbf{L}\mathbf{L}'$

$$\mathbf{L}\Phi = \mathbf{C}$$

Vector of i.i.d.
standard normal
variables $\rightarrow$

Vector of normal
variables that
satisfy Σ $\leftarrow$

Mathematics & Statistics for Financial Risk Management

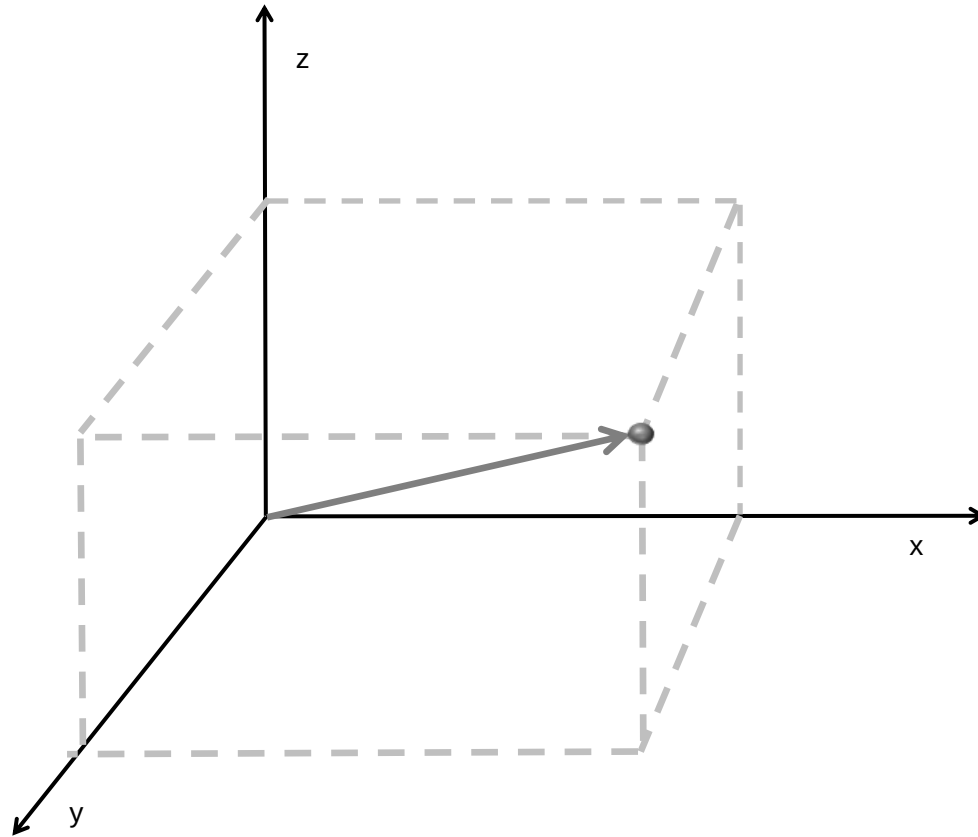
Topics



- Vectors revisited
- Orthogonality
- Rotation
- Principal Component Analysis
- The Dynamic Term Structure of Interest Rates
- The Structure of Global Equity Markets

Vectors Revisited

$$\mathbf{v}_2 = \begin{bmatrix} 5 \\ 10 \\ 4 \end{bmatrix}$$



Vectors Revisited

- Vector operations

- ▣ Inner product or dot product

$$\mathbf{a} \cdot \mathbf{b} = a_1 b_1 + a_2 b_2 + \dots + a_n b_n$$

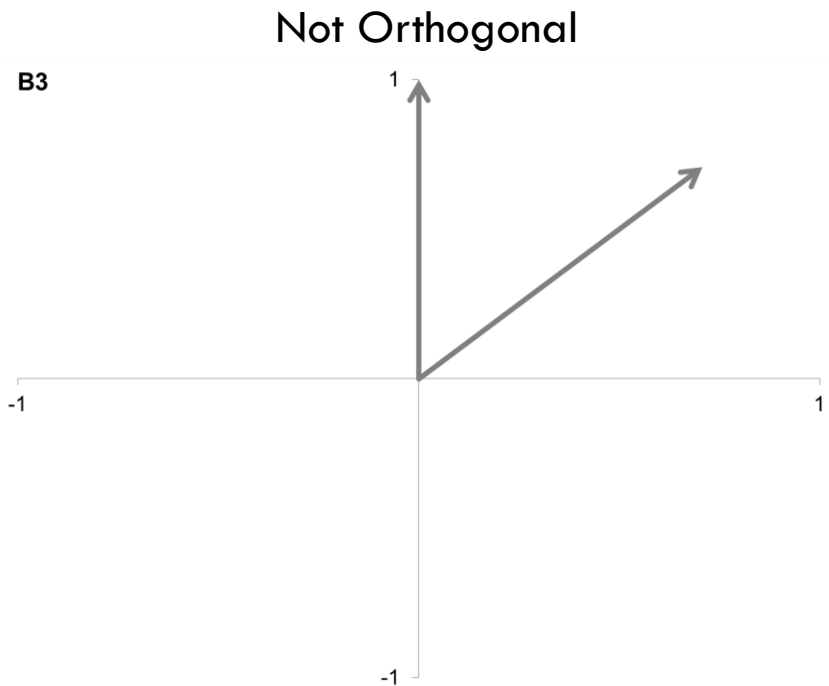
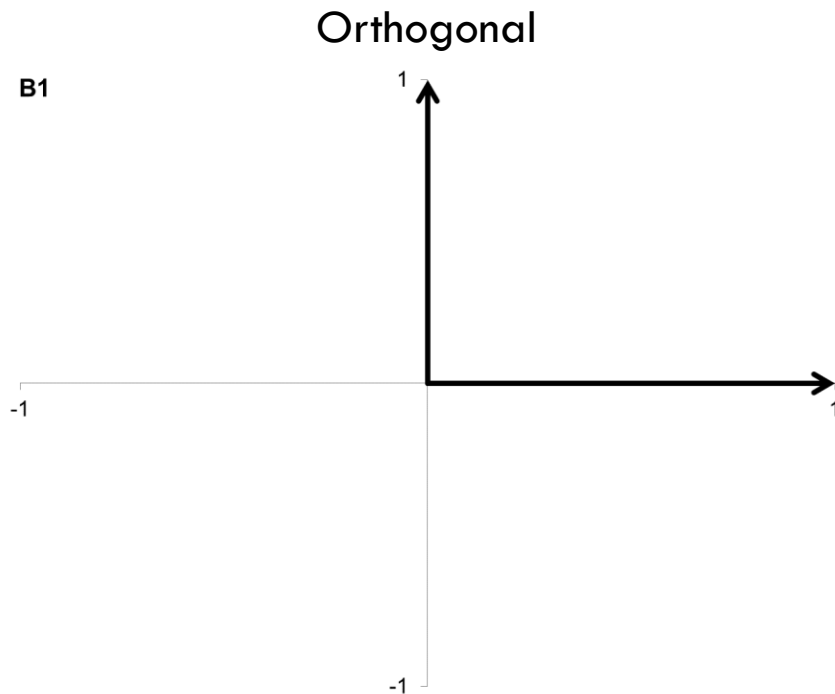
$$\mathbf{a} \cdot \mathbf{b} = \mathbf{a}' \mathbf{b}$$

- ▣ Length/norm/magnitude

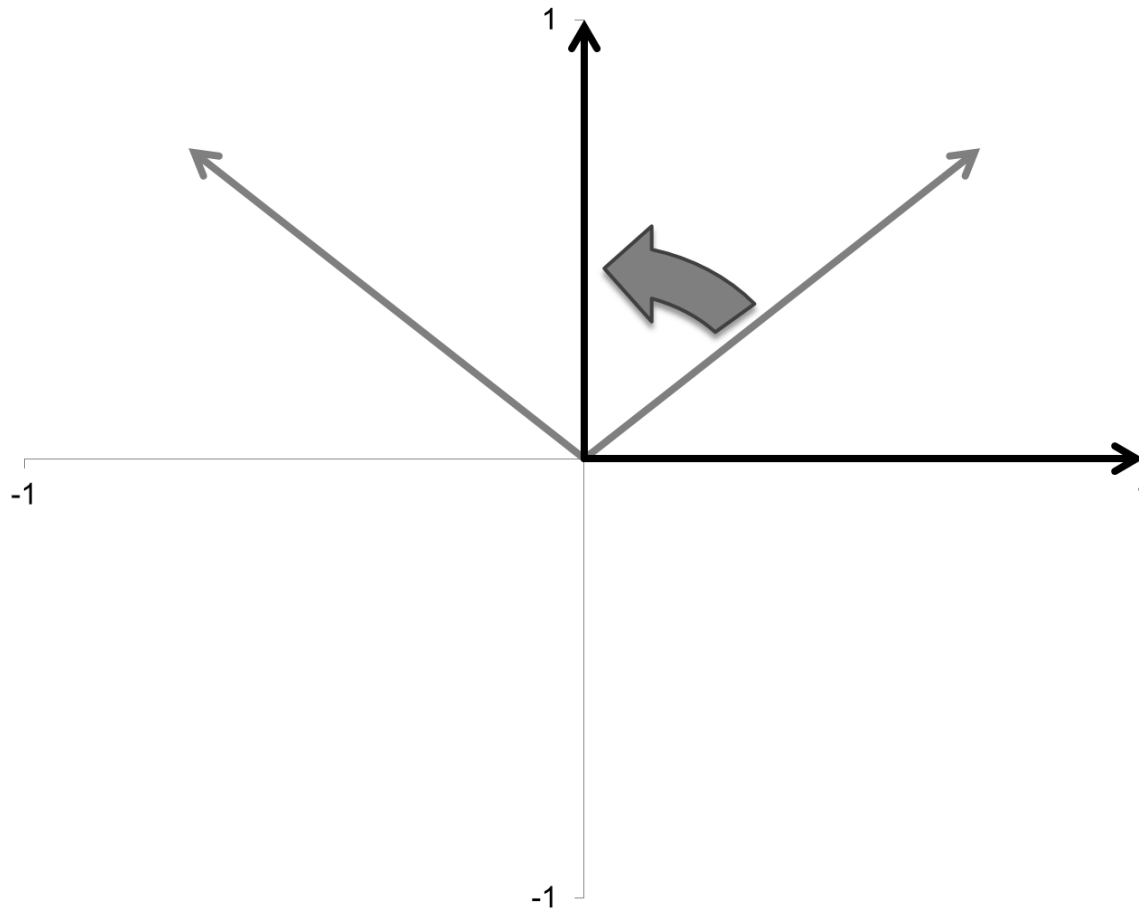
$$\|\mathbf{a}\| = \sqrt{\mathbf{a} \cdot \mathbf{a}}$$

Orthogonality

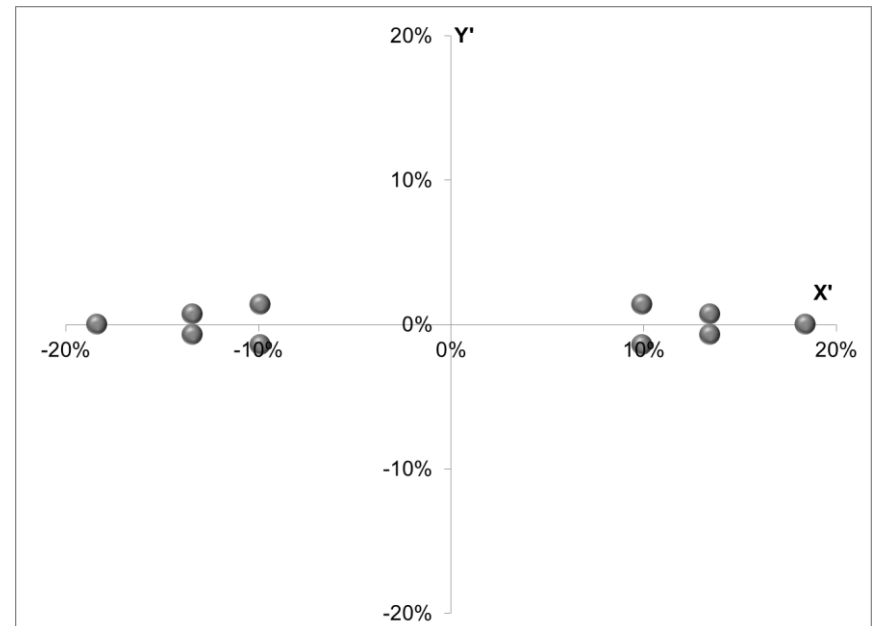
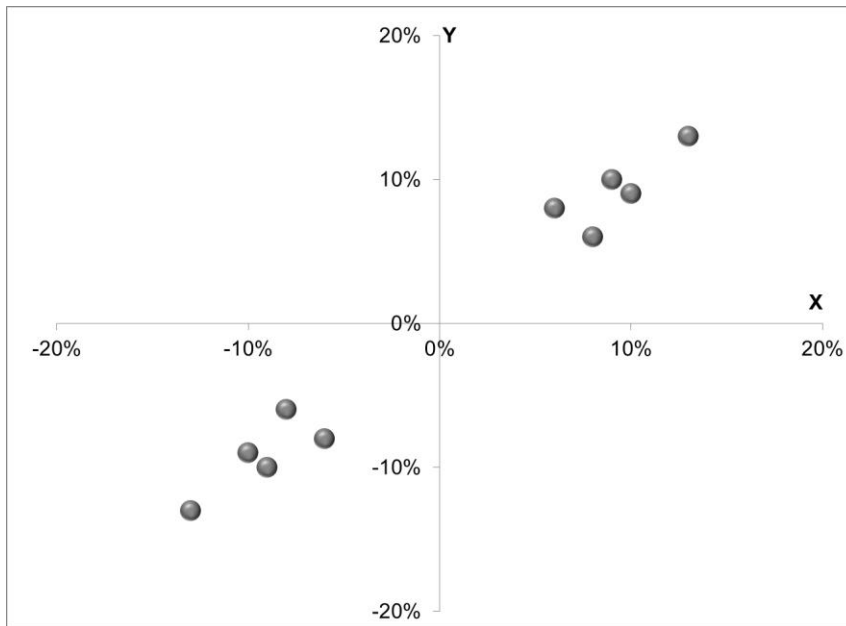
- Linear independence
- Inner product = 0



Rotation

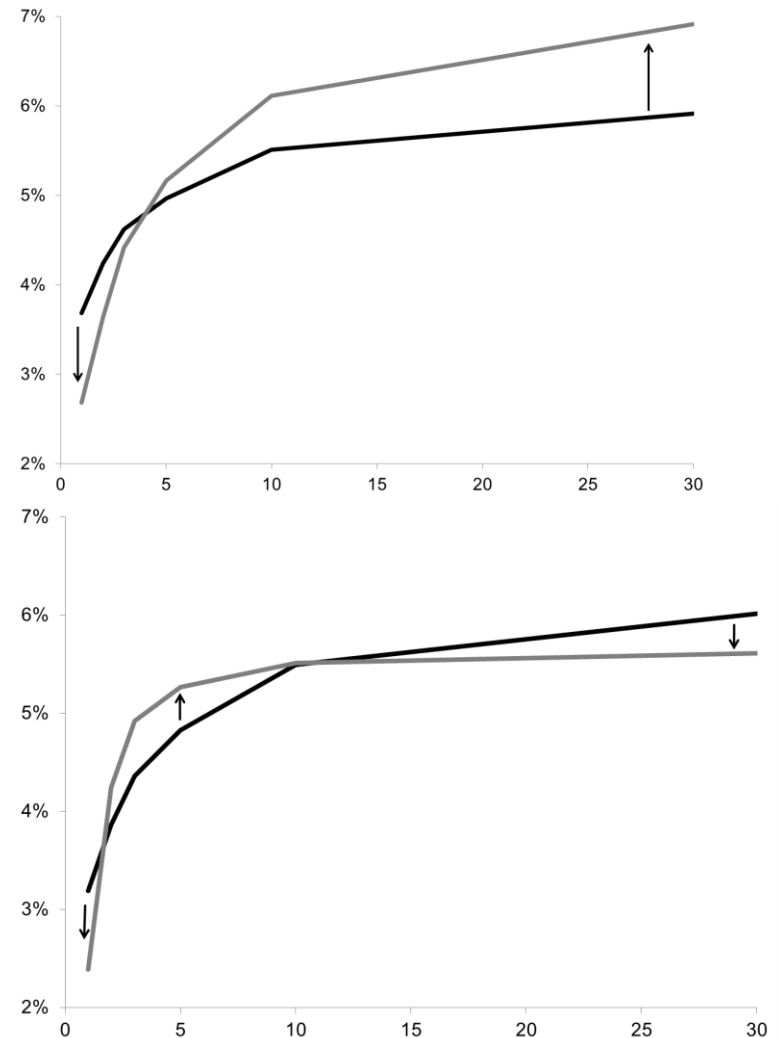
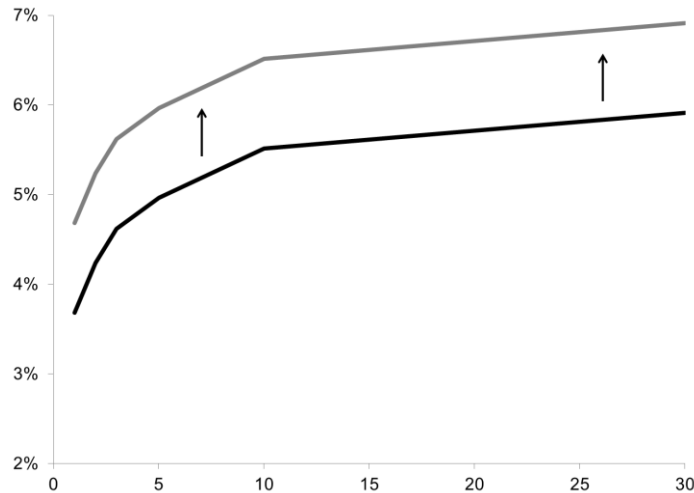


Principal Component Analysis



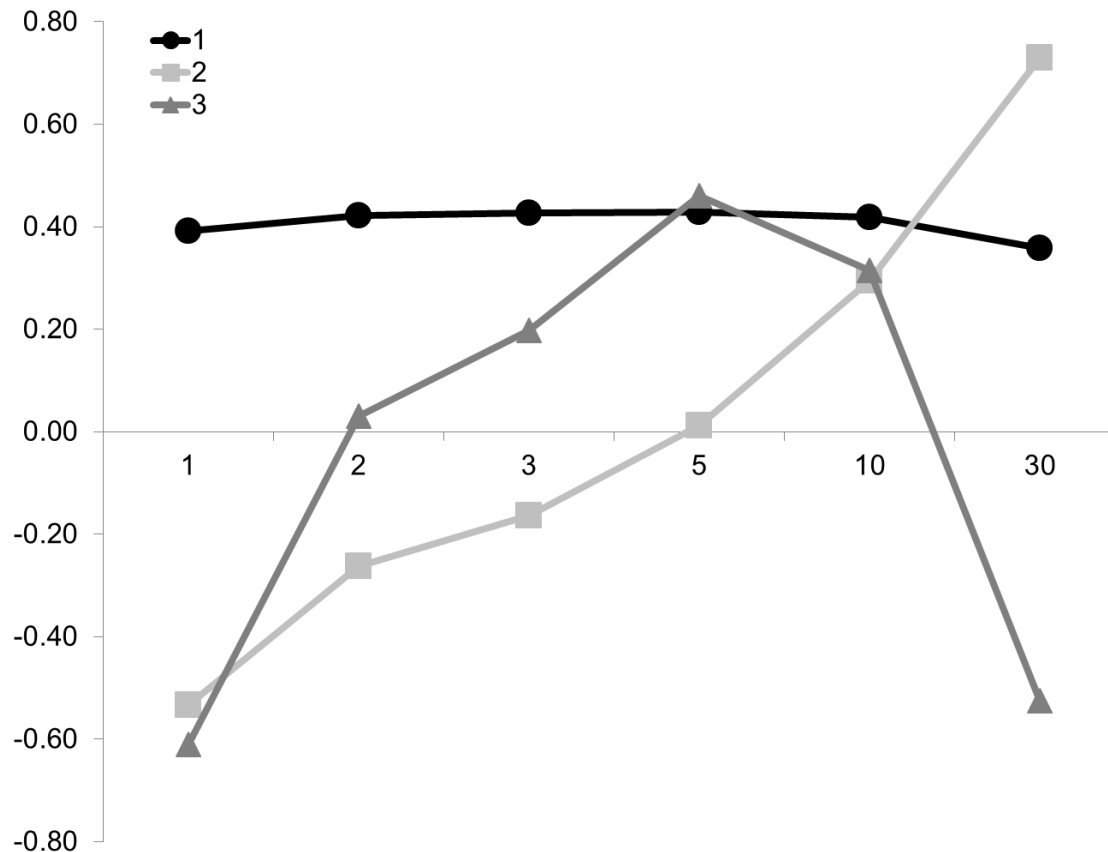
The Dynamic Term Structure of Interest Rates

□ Shift, Tilt and Twist



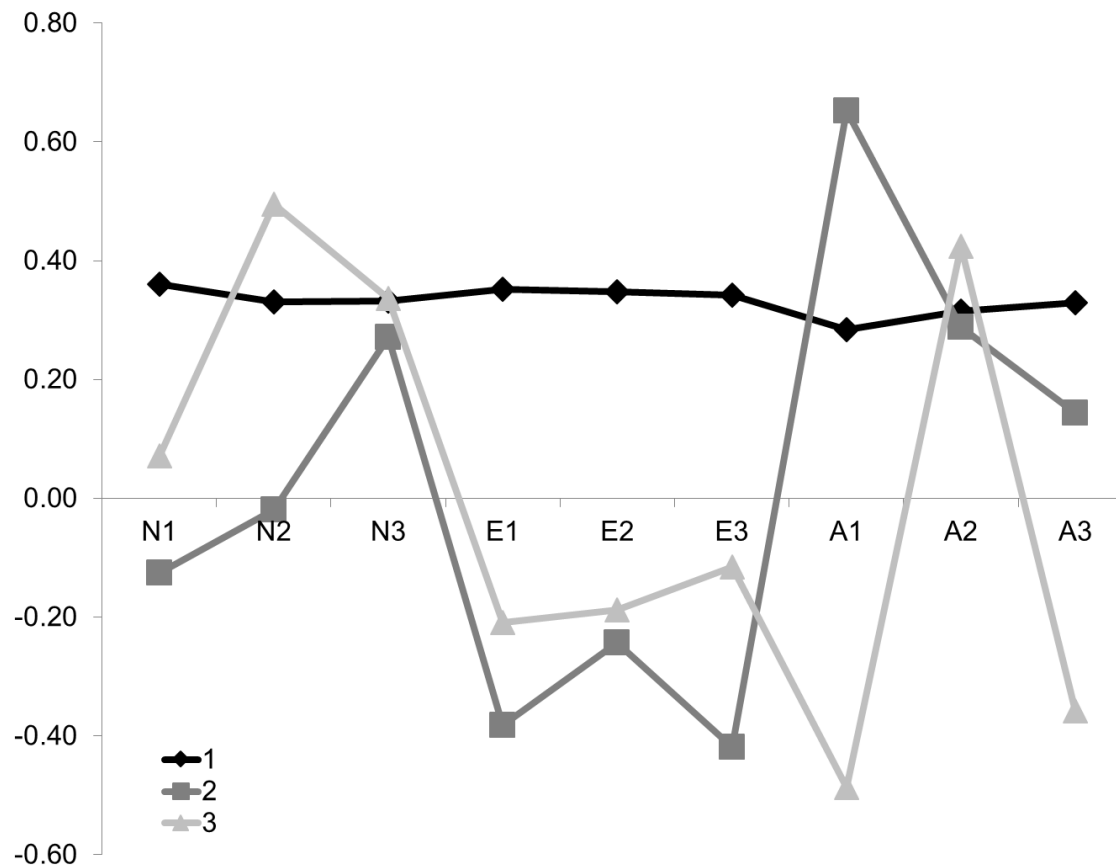
The Dynamic Term Structure of Interest Rates

- First three principal components of yield curve



The Structure of Global Equity Markets

- First three principal components of equity indexes



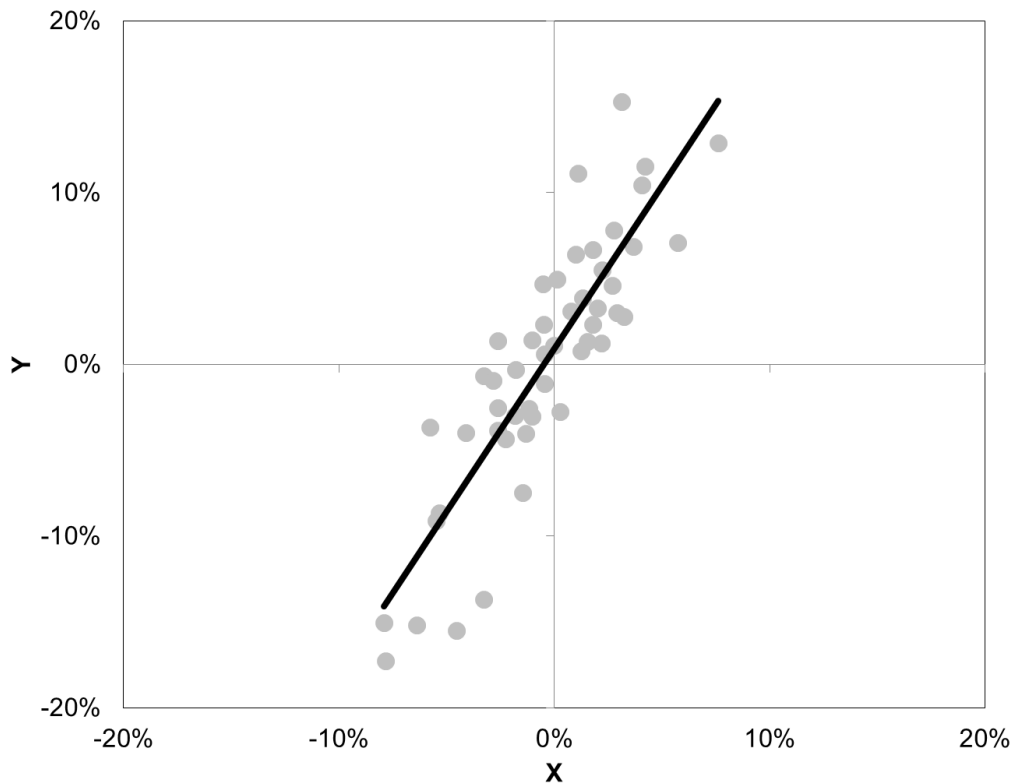
Mathematics & Statistics for Financial Risk Management

10: Linear Regression Analysis

Topics

- Linear Regression (one regressor)
- Linear Regression (multivariate)
- Factor Analysis
- Stress Testing

Linear Regression (one regressor)



$$Y = \underbrace{\alpha + \beta X}_{\text{systematic}} + \underbrace{\varepsilon}_{\text{idiosyncratic}}$$

Linear Regression (one regressor)

- Ordinary Least Squares (OLS)
 - ▣ The relationship between the regressor and the regressand is linear.
 - ▣ $E[\varepsilon | X] = 0$
 - ▣ $\text{Var}[\varepsilon | X] = \sigma^2$
 - ▣ $\text{Cov}[\varepsilon_i, \varepsilon_j] = 0$ for all $i \neq j$
 - ▣ $\varepsilon_i \sim N(0, \sigma^2)$ for all ε_i
 - ▣ The regressor is nonstochastic.

Linear Regression (one regressor)

- Estimating parameters

- ▣ OLS seeks to minimize RSS:

$$\text{RSS} = \sum_{i=1}^n \varepsilon_i^2 = \sum_{i=1}^n (y_i - \alpha - \beta x_i)^2$$

- ▣ Solution

$$\alpha = \bar{Y} - \beta \bar{X}$$

$$\beta = \frac{\sum_{i=1}^n x_i y_i - n \bar{Y} \bar{X}}{\sum_{i=1}^n x_i^2 - n \bar{X}^2}$$

Linear Regression (one regressor)

□ Optimal hedging revisited

$$h^* = -\rho_{AB} \frac{\sigma_A}{\sigma_B} \quad \leftarrow \text{Result from Chapter 3}$$

$$\beta = \frac{\text{Cov}[X, Y]}{\sigma_X^2} = \rho_{XY} \frac{\sigma_Y}{\sigma_X} \quad \leftarrow \text{From on OLS assumptions}$$

$$\begin{aligned} r_A &= \alpha + \beta r_B + \varepsilon \\ h^* &= -\beta \end{aligned} \quad \leftarrow h^* \text{ negates systemic risk}$$

Linear Regression (multivariate)

$$Y = \beta_1 + \beta_2 X_2 + \beta_3 X_3 + \dots + \beta_n X_n$$

$$\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$$

$$\hat{\boldsymbol{\beta}} = (\mathbf{X}' \mathbf{X})^{-1} \mathbf{X}' \mathbf{Y}$$

Factor Analysis

- Divide risk into systematics and idiosyncratic components

$$r_{\text{portfolio}} = \alpha + \beta r_{\text{index}} + \varepsilon$$

$$\sigma_{\text{portfolio}}^2 = \beta^2 \sigma_{\text{index}}^2 + \sigma_{\varepsilon}^2$$

β , the factor exposure, can be added across securities.

Stress Testing

- Factors can be used to construct stress tests
- Potential problem if linear relationship does not hold for extreme events

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11: Time Series Models

Topics

- Random Walks
- Drift-Diffusion
- Autoregression
- Variance and Autocorrelation
- Stationarity
- Continuous Models
- GARCH
- Jump-Diffusion
- Interest Rate Models

Random Walks

□ Definition:

$$x_t = x_{t-1} + \varepsilon_t$$

$$E[\varepsilon_t] = 0$$

$$E[\varepsilon_t^2] = \sigma^2$$

$$E[\varepsilon_t \varepsilon_s] = 0 \quad \forall s \neq t$$

□ Alternate formulation:

$$\Delta x_t = x_t - x_{t-1} = \varepsilon_t$$

□ Properties:

$$E[x_t | x_0] = x_0$$

$$\text{Var}[x_t | x_0] = t\sigma^2$$

Drift-Diffusion

□ Definition:

$$p_t = \alpha + p_{t-1} + \varepsilon_t$$

□ Alternate formulation:

$$r_t = \Delta p_t = \alpha + \varepsilon_t$$

□ Properties:

$$\begin{aligned} E[p_t|p_0] &= p_0 + t\alpha \\ \text{Var}[p_t|p_0] &= t\sigma^2 \end{aligned}$$

Autoregression

□ AR1 model

$$r_t = \alpha + \lambda r_{t-1} + \varepsilon_t$$

$$r_t = \alpha \sum_{i=0}^{n-1} \lambda^i + \lambda^n r_{t-n} + \sum_{i=0}^{n-1} \lambda^i \varepsilon_{t-i}$$

$$E[r_t | r_{t-n}] = \frac{1 - \lambda^n}{1 - \lambda} \alpha + \lambda^n r_{t-n}$$

$$\text{Var}[r_t | r_{t-n}] = \frac{1 - \lambda^{2n}}{1 - \lambda^2} \sigma^2$$

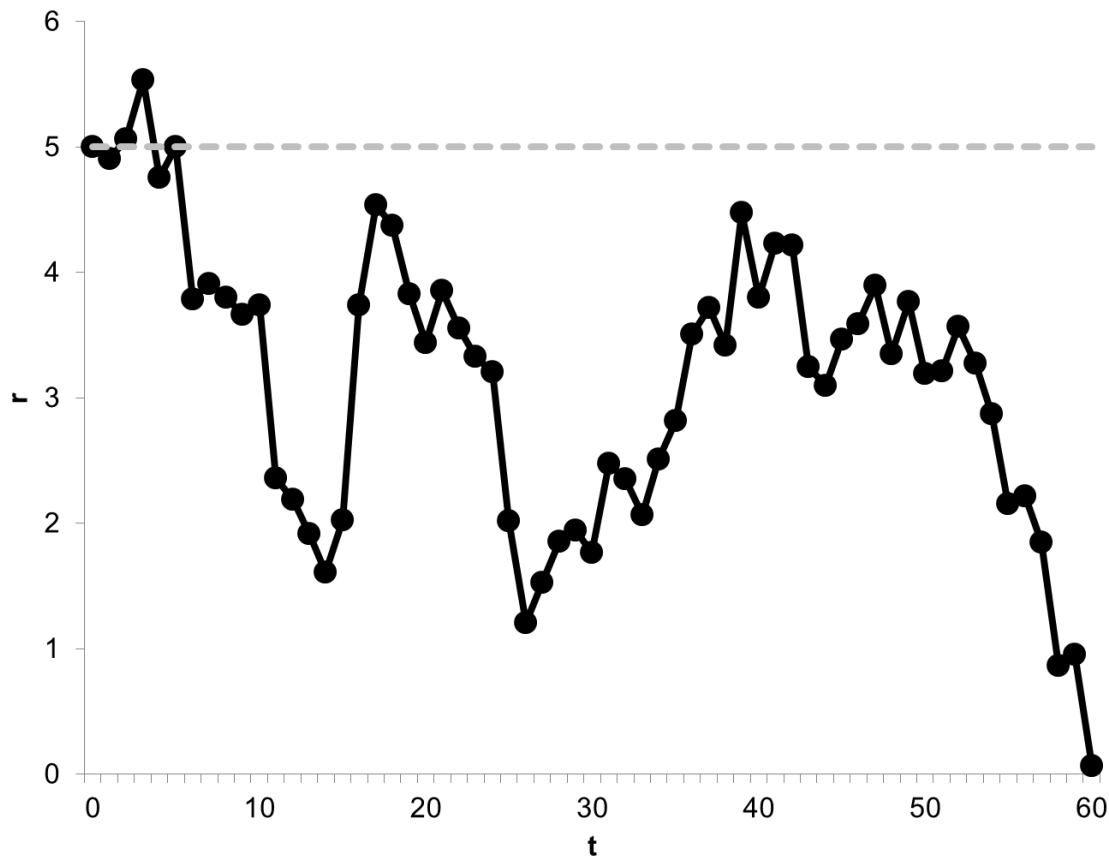
Autoregression

- Mean reversion ($\alpha = 0.50$, $\lambda = 0.90$)



Autoregression

- Divergent series ($\alpha = -0.10$, $\lambda = 1.02$, $\sigma = 0.50$)



Variance and Autocorrelation

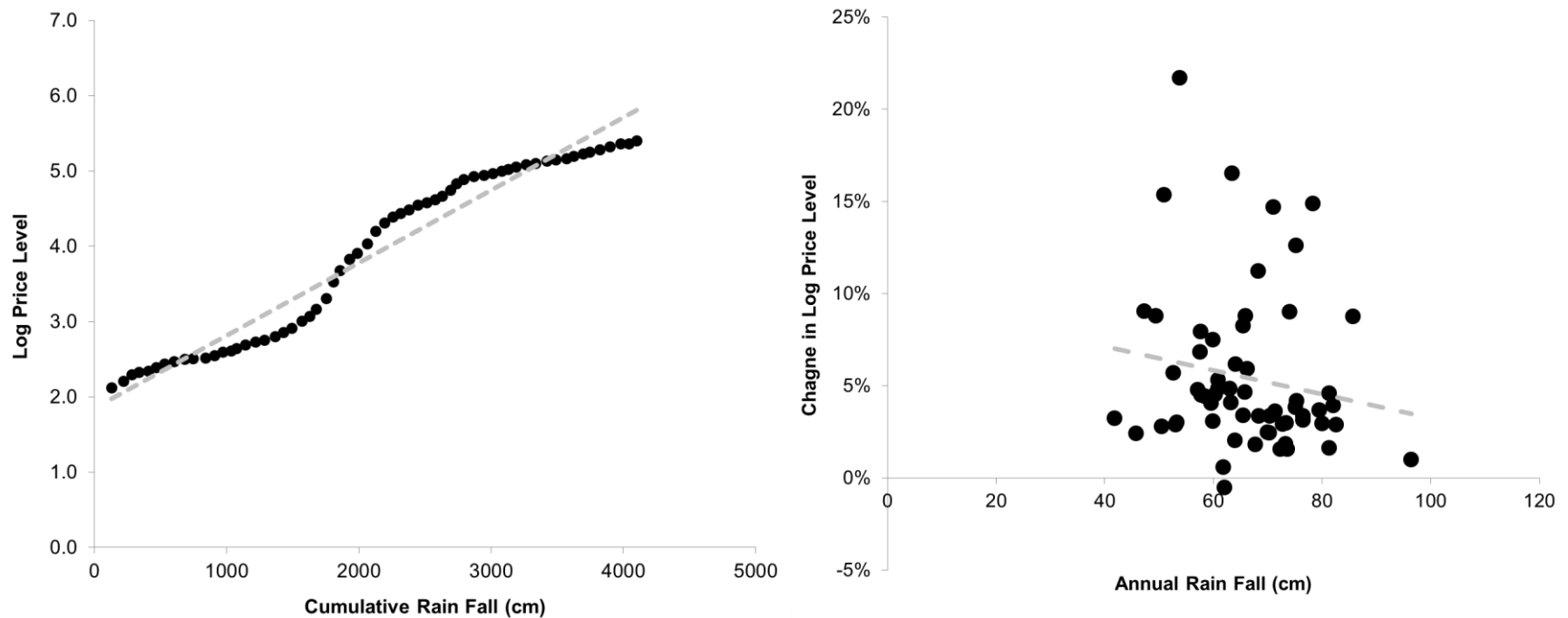
□ Given:

$$y_{n,t} = \sum_{i=0}^{n-1} r_{t-i} = \sum_{i=0}^{n-1} \varepsilon_{t-i}$$

□ For random walk: $\text{Var}[y_{n,t}] = n\sigma_{\varepsilon}^2$

□ With autocorrelation: $\text{Var}[y_{n,t}] = \frac{2}{1-\lambda} \sigma_{\varepsilon}^2$

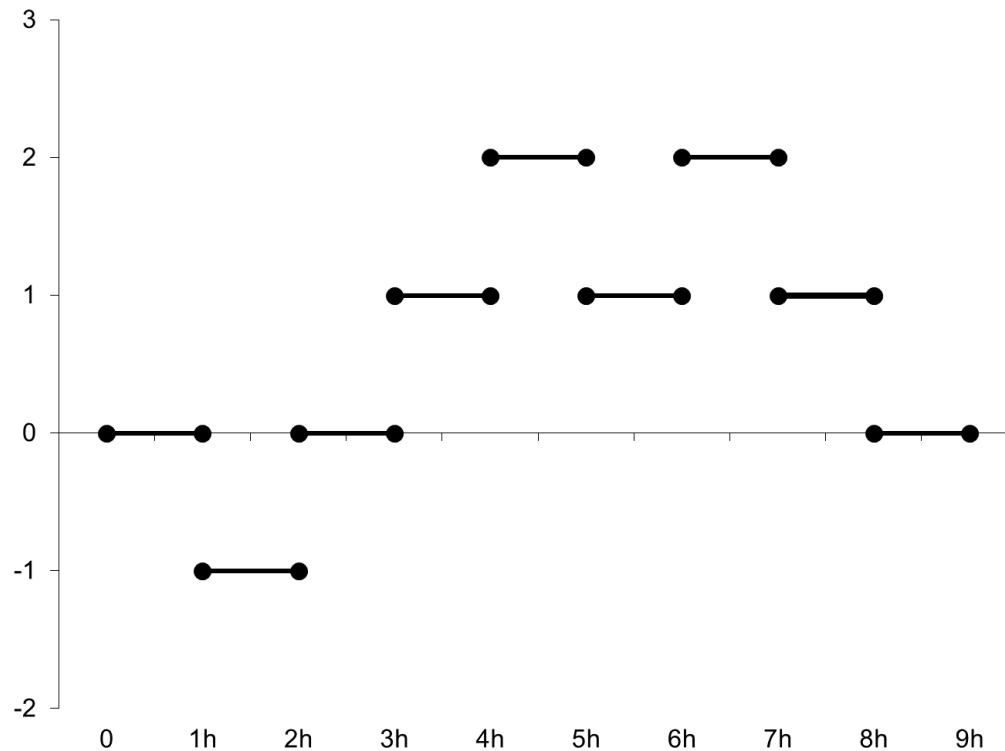
Stationarity



United Kingdom: Price Level and Rain Fall, 1949–2010

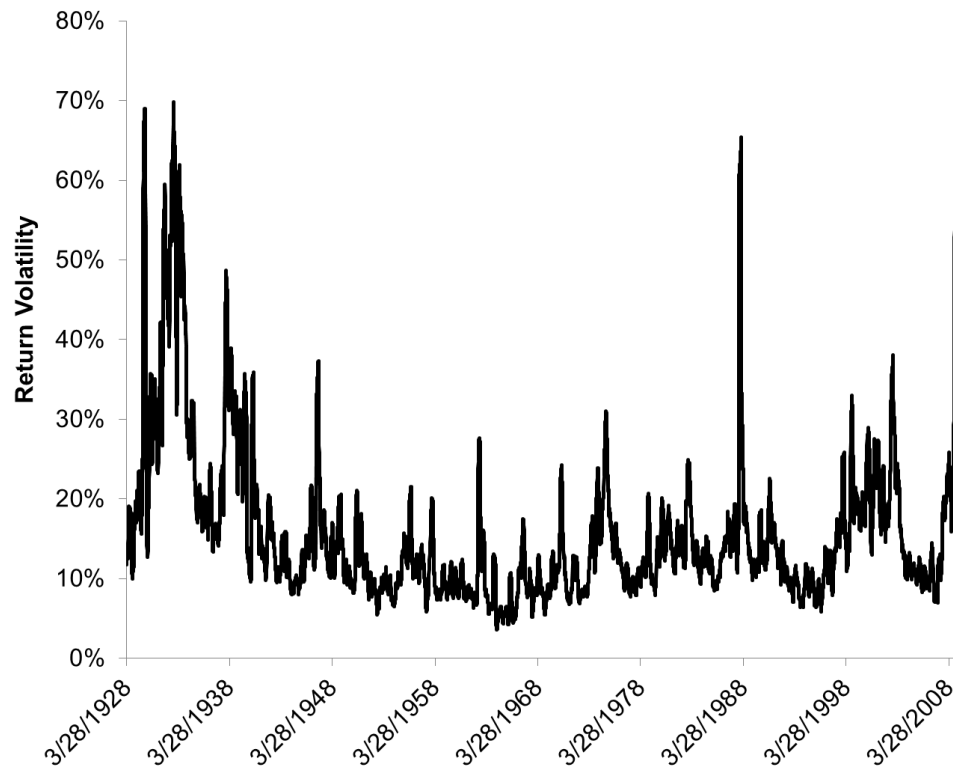
Continuous Models

- Discrete time series with infinitely short time interval



GARCH

- Markets do not appear to be homoscedastic



S&P 500, Annualized 60-Day Return Standard Deviation, 1928–2008

GARCH

□ ARCH(1)

$$\sigma_t^2 = \alpha_0 \bar{\sigma}^2 + \alpha_1 \sigma_{t-1}^2 u_{t-1}^2 = \alpha_0 \bar{\sigma}^2 + \alpha_1 \varepsilon_{t-1}^2$$

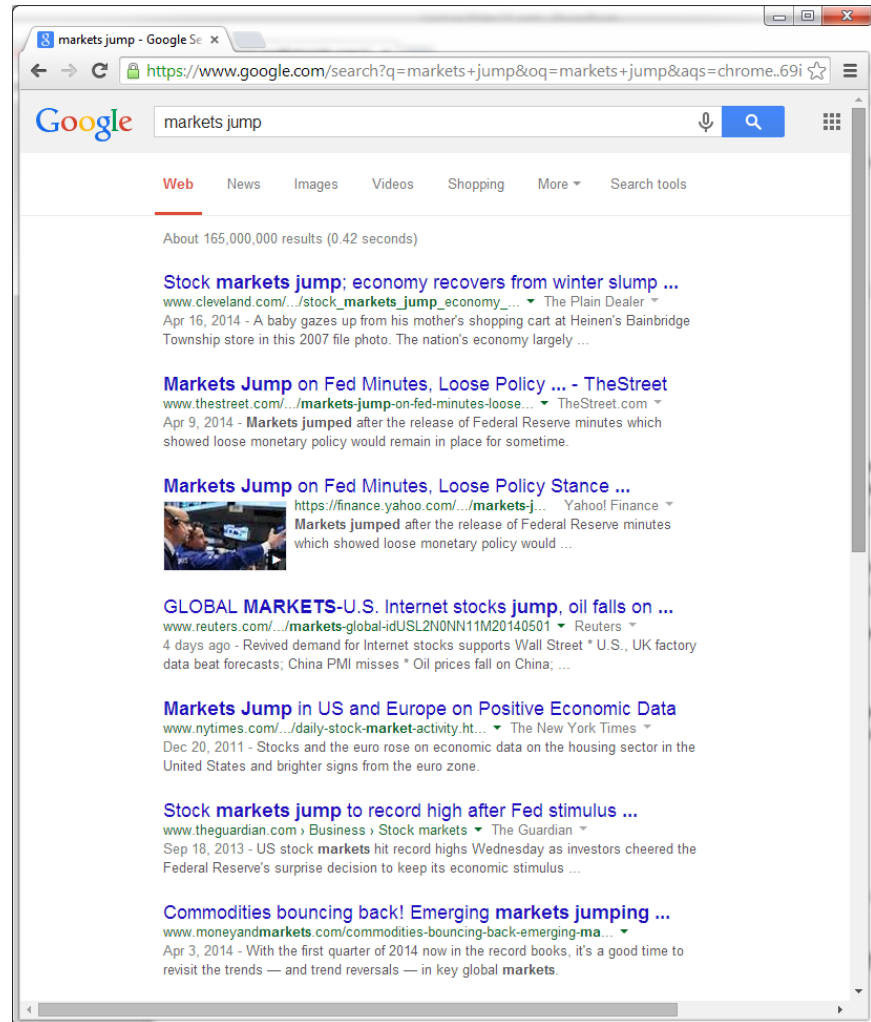
□ GARCH(1,1)

$$\sigma_t^2 = \alpha_0 \bar{\sigma}^2 + \alpha_1 \sigma_{t-1}^2 u_{t-1}^2 + \beta \sigma_{t-1}^2$$

Jump-Diffusion

- Markets tend to jump.

$$r_t = \alpha + \varepsilon_t + [I_t]u_t$$



Interest Rate Models

□ Vasicek

$$r_t = \lambda r_{t-1} + (1 - \lambda)\theta + \sigma r_{t-1}^{1/2} \varepsilon_t \quad \varepsilon \sim N(0,1), 0 \leq \lambda \leq 1$$

□ Cox-Ingersoll-Ross

$$r_t = \lambda r_{t-1} + (1 - \lambda)\theta + \sigma r_{t-1}^{1/2} \varepsilon_t \quad \varepsilon \sim N(0,1), 0 \leq \lambda \leq 1$$

□ CEV

$$r_t = \lambda r_{t-1} + (1 - \lambda)\theta + \sigma r_{t-1}^{\phi} \varepsilon_t \quad \varepsilon \sim N(0,1), 0 \leq \lambda \leq 1$$

□ Black-Karasinski

$$\ln(r_t) = \lambda \ln(r_{t-1}) + (1 - \lambda)\theta + \sigma \varepsilon_t \quad \varepsilon \sim N(0,1), 0 \leq \lambda \leq 1$$

Mathematics & Statistics for Financial Risk Management

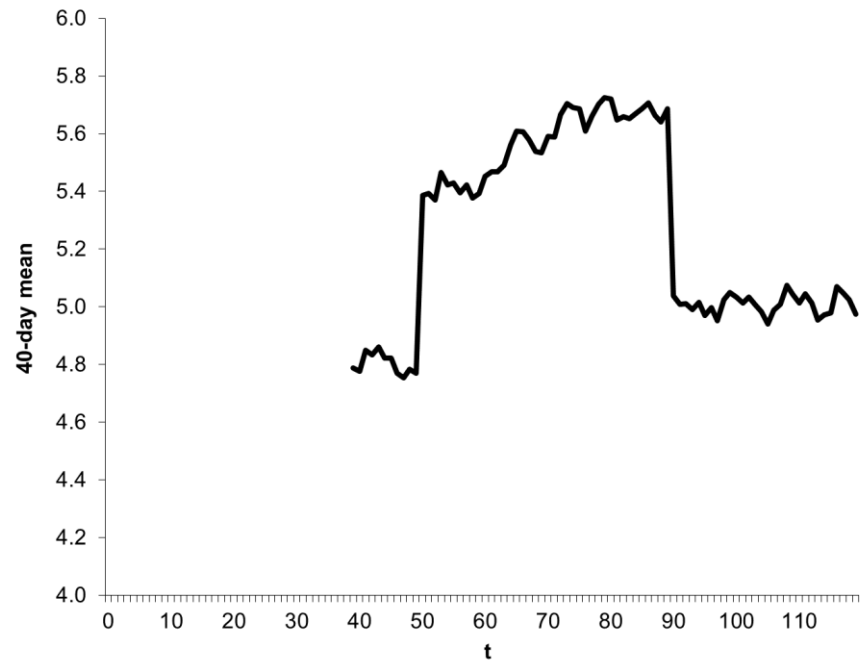
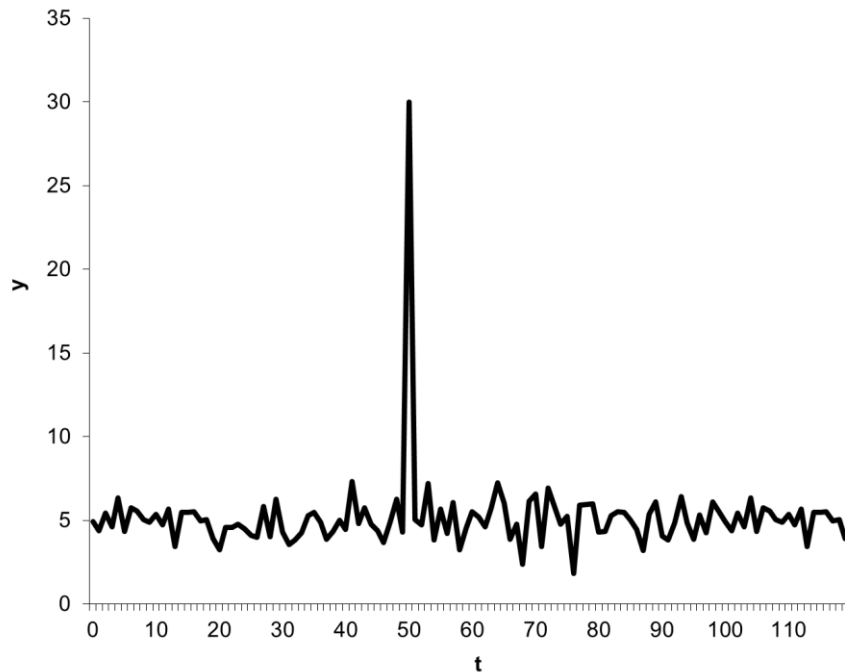
12: Decay Factors

Topics

- Mean
- Half-Life
- Variance
- Weighted Least Squares
- Hybrid VaR

Mean

- The problem:
Jumps lead to spikes, lead to plateaus.



Mean

□ A possible solution:

▣ Rather than:

$$\hat{\mu} = \frac{1}{n} \sum_{i=0}^{n-1} x_{t-i}$$

▣ Use:

$$\hat{\mu}_t = \frac{1-\delta}{1-\delta^n} \sum_{i=0}^{n-1} \delta^i x_{t-i}$$

$$\hat{\mu}_t = (1 - \delta)x_t + \delta\hat{\mu}_{t-1}$$

Half-Life

$$\sum_{i=0}^{h-1} \delta^i = \frac{1}{2} \sum_{i=0}^{n-1} \delta^i$$

$$h = \frac{\ln(0.5 + 0.5\delta^n)}{\ln(\delta)}$$

$$\frac{\text{weight of finite series}}{\text{weight of infinite series}} = \frac{1 - \delta^n / 1 - \delta}{1 / 1 - \delta} = 1 - \delta^n$$

Variance

- When mean is known

$$\hat{\sigma}_t^2 = \frac{1-\delta}{1-\delta^n} \sum_{i=0}^{n-1} \delta^i (r_{t-i} - \mu)^2 \quad 0 < \delta < 1$$

$$\hat{\sigma}_t^2 = (1-\delta)(r_t - \mu)^2 + \delta \hat{\sigma}_{t-1}^2$$

- When mean is unknown

$$\hat{\sigma}_t^2 = A \sum_{i=0}^{n-1} \delta^i r_{t-i}^2 - B \hat{\mu}_t^2$$

Weighted Least Squares

$$\tilde{\boldsymbol{\beta}} = (\mathbf{X}' \mathbf{W} \mathbf{W} \mathbf{X})^{-1} \mathbf{X}' \mathbf{W} \mathbf{W} \mathbf{Y}$$

$$\mathbf{W} = \begin{bmatrix} \lambda^{n-1} & \dots & 0 & 0 \\ \vdots & \ddots & 0 & 0 \\ 0 & 0 & \lambda & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad \lambda = \sqrt{\delta}$$

Hybrid VaR

- Decay factor is easy to apply to parametric VaR models (variance with decay could be used as a starting point).
- Historical VaR is the most basic non-parametric model (advantage: reproduces skew, fat-tails, serial correlation, GARCH, ...)
- Hybrid is historical VaR with decay factor